Mathematical Materiality 2.32: An AI-Assisted Cosmological Model Survives Its First Joint Test Across Three Cosmological Probes

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Abstract

A model of the universe developed with the assistance of GPT-6 Astra and Claude has survived its first global confrontation with cosmological data. Across supernovae, DESI measurements and the geometry of the cosmic microwave background, it obtains almost the same fit as the standard cosmological model: a chi-square of 1420.02, compared with 1421.19 for ΛCDM. This is not a statistical victory over ΛCDM. The news is that the construction did not collapse. Starting from a fundamentally different account of cosmic acceleration, Mathematical Materiality 2.32 reached almost the same agreement with three major families of observations. From four open problems to a formalized ontology Mathematical Materiality did not begin as an isolated attempt to explain dark energy. The proposal grew out of four problems that contemporary physics describes with great precision but still leaves open at a fundamental level:

A model of the universe, helped by AI, passes its first comparison with three kinds of observations

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Dark energy from growing relations

A Mathematical Materiality proposal explained step by step. The explanation uses examples, observations and formulas. The parts can also be read separately.

A cosmological model describes how the universe changes. This proposal starts from an unusual idea. Dark energy could be the effect of a network of relations that keeps growing. Dark energy is the name given to an effect. This effect seems to accelerate the expansion of the universe. We do not see it directly. We study its effects.

A growth rule sets how this component changes over time. The rule is controlled by three dark-sector parameters. A parameter is a number that can be varied in the model. The comparison uses 1,590 Pantheon+ supernova light curves. It also uses 13 DESI DR2 measurements and the summary distances of Planck 2018. Supernovae are very bright stellar explosions. Their light helps to estimate distances. DESI DR2 provides measurements linked to the distribution of matter in the universe. Planck 2018 summarizes some geometric properties of the cosmic microwave background. It is the very old light that fills the sky.

The chain of sampled values reaches a total chi-square of 1420.02. The chi-square summarizes how much the calculations and the measurements differ. It takes the uncertainties into account. With the same data and procedure, a lower value indicates a closer agreement. In the same calculation, a cosmological constant reaches 1421.19. The cosmological constant is a component with constant density. It is used as a reference for dark energy. The parameter p

measures how far the model departs from that reference. It has an upper bound of 0.0724 at 95 per cent credibility. In simple words, according to this analysis the values compatible with the model lie below that bound. The credibility is 95 per cent. It is a bound, not a precise measurement of p.

The model has three extra parameters in the dark sector. The result shows that it can agree with the data. It does not show that it is preferred. There are statistical criteria that also take the number of parameters into account. These criteria favour the cosmological constant. Moreover, the cosmic microwave background enters only through its geometric summaries. It does not enter through all the temperature and polarization data. The archived materials allow independent checks. A kit to redo the calculations with public data is planned in a separate record.

Figure 1. The path the text will follow. The upper row presents the proposed rules on relations. The lower row introduces their physical meaning. Then it leads to the observable measurements. The change of colour marks a change in the assumptions, not only in the words.

Why this proposal may be important

Question	Answer
What is new?	An expansion history that follows from the growth of a network of relations, without leaving a function of the redshift free. Redshift indicates how much the light has been stretched during its journey.
What happened?	The model was compared together with Pantheon+, DESI DR2 and the geometry of the cosmic microwave background.
Did it beat the cosmological constant?	No. The dark sector uses three extra parameters and the model-selection criteria favour the cosmological constant.
What is needed now?	That independent groups redo the comparison, and that all the cosmic microwave background data be used, instead of only the geometric summaries.

When we look at a distant galaxy, we receive light that set out a long time ago. That light carries an image and traces of its journey. The traces also depend on how the universe has expanded. We can measure them and compare them with a theory. A theory is a set of ideas and rules that make predictions. These measurements are now precise. It is much less clear what dark energy really is.

Mathematical Materiality, or MM, approaches the question from a different starting point. Before asking what substance fills space, it asks another question. What makes objects, the differences between them and their relations possible? In this proposal, the fundamental element is the **relational link**. It is a connection that makes it possible to relate distinguishable states. These states are like two different signs on a blackboard.

This text follows a single line of inquiry. Can an organized growth of relations become a component that accompanies cosmic expansion? Can this component also support the acceleration of the expansion? The answer developed here is an explicit model, compared with observations. The text also shows where counting ends and where added physics begins.

Independent cosmologists are invited to redo the comparison or to show that the result does not hold. The archived materials include the settings of the comparison, the sampled calculations and the checks carried out. A separate kit is also planned. The kit recalculates the main numbers with public data and the same calculations.

What the comparison shows and what it puts to the test

Aspect	What results from the comparison
Fit to the data	The chain of sampled values reaches a total chi-square of 1420.02. The minimum calculated separately for the profile is 1420.05. The difference is numerical, not physical. The cosmological constant, calculated with the same code and the same data, reaches 1421.19.
Cost of the model	The model uses three extra parameters in the dark sector. The improvement of about one unit of chi-square does not offset the additional parameters. So the selection criteria favour the cosmological constant.
Bound	The parameter p, which controls how far the model departs from the cosmological constant, stays below 0.0724 at 95 per cent credibility. It is an upper limit, not a measurement.

Result that would refute it	The growth rule forbids the ratio w of the dark component to fall below -1. A measurement that established this crossing would exclude this form of the model.
Next test	The non-uniform part of the model, described in advance in Appendix V, with a decision rule fixed before the calculation.

Appendices S-V report the complete comparison. Three kinds of geometric observations were compared together. The ordinary cosmological parameters were left free to adjust. Among these are the expansion rate and the amount of matter. The data include 1,590 Pantheon+ supernova light curves. They also include thirteen measurements of baryon acoustic oscillations. Then there are three summary numbers for the geometry of the last scattering of light. Baryon acoustic oscillations are a characteristic trace in the distribution of matter. Last scattering is the epoch in which the ancient light began to travel freely through the universe.

The ordinary cosmological parameters came out of the comparison. They were not imposed beforehand. The expansion rate and the matter density both turn out within about one standard deviation of the Planck values. The standard deviation indicates how much the values can vary around an estimate. This comparison assumes the standard cosmological framework for the expansion and for the formation of the first light we observe.

A direct calculation of the profile, without the emulator, finds the minimum of the chain again. An emulator is an approximate calculator. It reproduces longer calculations, after being checked. A sampled chain is a sequence of many choices of parameters. It serves to explore which values agree with the data. Two other chains give compatible bounds. These chains did not reach the final stability criterion. Two numbers remain free. They set when the departure from the cosmological constant begins and how quickly it switches on.

Now the construction of the model begins. We will see what a relational link is and how counting becomes a history of the expansion. We will also see where physics is added. Those who want to see the comparison first can go directly to Appendix S.

The question that comes before objects

Let us imagine we want to describe a city. We can list the buildings, the streets and the distribution of the inhabitants. We can also ask which relations make that form of organization possible. For example, which connections allow people to move, communicate and build something that lasts? These are two levels of description. The second does not erase the first. But it changes the point from which we start.

MM tries to do something similar for physical reality. The word *ontology* indicates the question about what we regard as fundamental. The question also concerns what could arise from its organization. In MM, objects and observers are described as configurations of the same network of relations. The reference point used to describe something is also part of the world being described.

This work grew out of four questions. Why do physical laws keep their regularity in different places and epochs? What lies behind the phenomena attributed to dark matter? What mechanism might account for dark energy? What picture of reality could make quantum behaviour intelligible?

Here we address only the third question. Dark matter and dark energy have similar names. But they refer to different problems. The *distal* component studied in these pages concerns the expansion of the universe. It is not an attempt to account for the rotation curves of galaxies. Rotation curves show the way stars orbit inside galaxies. A result obtained for one component is not transferred automatically to the other.

The image of the city helps to distinguish levels of description. But it is not physical evidence. In the same way, starting from relations is not enough to obtain a theory. Rules are needed that say which states are possible. They must also say how the states change and what we can calculate. Only then can we check whether the idea leads to a history of the universe consistent with observations.

Our aim is concrete. We want to build a path from the relational link to a history of the expansion. This history must be comparable with observations. Every necessary step must be visible. The result will not rest only on the evocative effect of the word "relation".

A relational link, explained without mysteries

To begin, two *registers* are enough. Let us imagine them as two boxes. Each can show a positive or a negative sign. They are small mathematical systems, not necessarily physical objects. Here "recording" means only having a state

distinguishable from another. Let us think of two squares on a blackboard with different signs.

We connect the boxes with a link. We want their states not to be completely independent. But they must not be forced always to agree either. A simple possibility is to assign more internal cases to agreement than to disagreement.

Take, for example, a given first sign. Imagine that the link has three ways to preserve it and one way to reverse it. If the four cases are equally likely, the second register repeats the first three times out of four. The difference between the cases of agreement and those of disagreement is three quarters minus one quarter: one half.

We call q the transmitted share of the correlation. In the example, $q = 1/2$. Careful: the probability that the signs agree is $3/4$. The coefficient q instead is $1/2$. Here we adopt a symmetric distribution. With this distribution, q measures how much the initial sign remains recognizable. The comparison is with a completely random choice.

If the cases of agreement and disagreement were equal in number, the second register would give no information about the first. If disagreement were impossible, the sign would pass through without attenuation. The model we will use lies between these extremes. The transmitted share is positive and smaller than one.

We have not yet introduced metres, seconds or masses. We have not introduced quantum entanglement either. It is the particular connection studied in quantum mechanics. This is a classical probabilistic channel. Its task is more precise and narrower. It must give a mathematical form to the "fractionating link". It does so through a finite count of possibilities.

At this level, the observer is not an external consciousness. One register can be compared with another and serve as a reference. Building a complex material observer is another question. The mechanism described here works even without someone watching it.

What changes when a third register is inserted

Now let us insert a box between the first two. The first link transmits half of the correlation. The second transmits half of that. At the end a quarter remains: half of a half is a quarter.

This is the operational meaning of *interposed fractionation*. The comparison between the endpoints passes through distinct steps. Each of these steps contributes to the result. With two links we write $Q = q_1 q_2$. In a longer chain we multiply all the transmitted shares. The capital letter denotes the product over the entire path.

We can turn this multiplication into an addition by using the logarithm. The logarithm is an operation that turns products into sums. The quantity $-\log Q$ grows when the retained correlation decreases. The contributions of the single steps add up along the chain. In this way we measure the depth of mediation, still without physical units. The precise proof is in Appendix A.

What have we gained? A distinction between direct comparison and mediated comparison. This distinction does not depend only on the names given to objects. We have not yet obtained a distance in space to be measured with a ruler.

An example clarifies the boundary. Suppose I draw a link as two segments instead of one. But I choose the fractions so that the total product stays the same. In this case the system has not changed. I have changed the drawing. It would be wrong to say that space grows every time I add a node to the drawing.

We therefore distinguish two operations. The first is describing the same relation more finely. The second is generating a new state of the network. Only the second changes the available capacity. This precaution makes the mechanism independent of arbitrary details of the drawing.

The product formula holds for the chain built here. In a network with several paths and cycles, the correlations can receive different contributions. Without a proof, we will not replace that problem with a formula valid for a single path.

Ten slots, eleven slots, twenty slots

Let us move from transmission to growth. We denote by n the number of *slots*. Slots are places where a state can be recorded. A simple rule replaces one slot with two at each update. The total therefore grows by one: ten, eleven, twelve and so on.

We want to connect this growth with fractionation. To do so we propose a precise channel between one state of the network and the next. If the network has n slots, we assign $2n + 1$ internal possibilities to agreement and one to disagreement. For example, with ten slots there are twenty-one cases of agreement and one of disagreement. The difference is divided by the total number of cases. The result is twenty divided by twenty-two, that is, ten elevenths.

$$q_n = \frac{n}{n+1}, \qquad Q_{n_i, n_f} = \frac{n_i}{n_f}. \qquad (1)$$

The second part of the formula can be understood with a simple calculation. Let us go from ten to twenty slots. We multiply ten elevenths, eleven twelfths, twelve thirteenths and so on, up to nineteen twentieths. The intermediate numbers cancel. Ten twentieths remain, that is, one half.

Capacity has doubled. The overall transmitted share has halved. In general:

$$C = \frac{n_f}{n_i} = Q^{-1}_{n_i, n_f}, \qquad u = \log C. \qquad (2)$$

Here C indicates how much the number of slots grows. It does not measure cubic metres. The quantity u counts that growth on a logarithmic scale. It is not yet the time shown by a clock.

[CANDIDATE MM RULE] We chose these counts to obtain an exact connection between fractionation and capacity. The construction shows that the connection is possible and consistent. It does not prove that MM's ontology must select exactly these counts. The dependence on the total number of slots is a collective rule. It remains to be explained how it could arise from a local cause.

It is also important to know where the chain lives. It connects successive states of the whole process under consideration. It does not say that every path in space between two galaxies has the same fraction. The relation between cosmological growth and observed distance will be built later.

The network can grow without preserving everything

Increasing the available slots does not mean increasing in the same proportion what remains active. Let us imagine an archive. Some new entries add a useful distinction. Others repeat information already reconstructed. The example separates total capacity from activity. It does not presuppose that a cosmic archivist exists.

We use three labels. A and B denote two active types. I denotes a slot that is inactive for the count we are tracking. "Inactive" does not mean nonexistent. That slot remains included in the total capacity.

At each step one slot is chosen at random. All slots have the same probability. If A is chosen, we replace it with A and B. If B is chosen, two B slots usually appear. With a certain probability, called p, A and I appear instead. If I is chosen, two I slots appear.

Figure 2. A proposed rule, not a process observed through a telescope. Each row replaces one slot with two. So capacity always grows by one. The active labels can increase or be redistributed. The probability p concerns only the choice between the two possible outcomes for type B.

The letters A and B do not denote known particles. They are minimal states of a possible mechanism. Their role depends on the operations they permit, not on the names chosen.

An explicit rule lets us follow the process. We do not have to decide in advance which cosmological result we would like to obtain. Once the rule is fixed, we can calculate a distribution of possible histories. We can find the mean growth, the random fluctuations and the response to changes in the rule.

Two probabilities not to be confused

The possibility that A and I are obtained from B is called *redundancy*. We denote it by p. A simple way to realize it is to draw two independent labels. The case in which they are equal is regarded as redundant.

Take four equally likely labels. There are sixteen possible pairs and four equal pairs. The probability is one quarter. In general, one sums the squares of the label probabilities. The number four is a choice in the example, not a cosmological

result.

Now we have two different quantities. The fraction q describes how much of the comparison passes from one state to the next. The probability p governs which records remain active while capacity grows. Calling both of them "fractions" would hide an essential difference.

The two constructions can coexist in a single probabilistic process. Appendix B gives the joint rule. It also checks that the probabilities sum to one. There is no need to pretend that one rule has been automatically deduced from the other.

The uniform choice also has a meaning. It says that a slot is not chosen more often merely because it carries the label A, B or I. If we changed this choice, the dynamics could change. Here *robustness* means that a behaviour is retained when specific changes are made to the rules. It does not mean that it is retained under every imaginable change.

This is a difference between a free narrative and a model. In a narrative we can give each word the behaviour that is needed. In a model we must accept the consequences of the stated rules. We must accept them even when they do not lead to the desired result.

The average of possible histories

Let us repeat the game many times starting from the same initial state. The sequences will be different. At one step A will be chosen, at another I. Chance will also affect the outcomes of B. We can nevertheless ask how many active registers there are on average after a given increase in capacity.

For this rule, the calculation of the mean closes exactly. The next step depends linearly on the current means. To write that particular equation we do not have to invent an approximate form of the correlations. The matrix in Appendix B collects in an orderly way the contributions of the two active types.

When the initial number of slots is large, we can describe growth continuously. This means replacing many small updates with a curve. The error is controlled in the specified limit. It does not mean that the network has become a continuous spacetime.

Let us consider the long run, in the model's dominant regime. The mean number of active registers grows as capacity raised to an exponent λ. An exponent says how quickly one quantity grows compared with another. In this case it is calculated like this:

$$\lambda = \frac{1 - 2p + \sqrt{1 + 4p^2}}{2}. \qquad (3)$$

There is no need to memorize the formula. Its message is this. Once the redundancy rule is fixed, the relative growth rate is no longer free. For $p = 1/4$, it is about 0.809.

A power with an exponent smaller than one grows more slowly than the starting quantity. Capacity can therefore keep increasing while the share of slots occupied by active registers decreases. At the same time, the total number of active registers can grow. The two things do not contradict each other.

The mean alone does not describe a single sequence. For this reason Appendix F also retains the calculation of the random fluctuations. It specifies the intervals in which their relative effect decreases when the initial number of slots grows.

More active registers, but less activity per slot

Let us imagine a numerical example in the dominant regime. We use the exponent just calculated. If capacity doubles, the active total is multiplied by about 1.75, not by two. So it grows by about 75 per cent. But now we divide the total by a doubled capacity. The active share per slot falls to about 88 per cent of its initial value.

This is the heart of the mechanism: **the total grows, while its concentration decreases slowly**. It is not yet a law of dark energy. It is the statistical behaviour that we will try to translate into physical language.

Figure 3. An example that uses only the dominant mode, with $p = 1/4$. It compares capacity, the normalized active

total and their ratio. All curves start at one to make reading easier. This does not mean that every initial slot is active. They are not astronomical data. Nor is it the complete curve used in the comparison with observations.

The difference can also be seen by imagining a capacity eight times larger than the initial one. The active total grows by about 5.38 times. Meanwhile its share of the capacity falls to about 0.672. If the quantity were merely conserved, without new active growth, the dilution would instead be eightfold.

Why might cosmology care? When a volume increases, what matters is how much the energy it contains changes. What matters is not only how much energy it contains. Two components with the same density today may have different histories. They may lead to different evolutions.

Before speaking of energy, however, we must cross a boundary. A count of records does not automatically have the units of a volume or of an amount of energy. This step must be stated, not left to the reader.

The step from counts to physics

On a map we can measure distances in centimetres. To obtain kilometres we need a scale. Here there is a deeper question. We must not only choose a unit. We must decide which physical property a count represents.

We propose two identifications. The first assigns to every slot the same volume contribution. More slots mean more volume for the region considered. The second assigns the same energy to every active record. It uses the mean number of active records. These are simple, testable rules. The fact that the link fractionates the correlation does not prove them.

[IMPORTED EFFECTIVE POSTULATE] Volume is proportional to capacity. Energy is proportional to the mean number of active records. Volume is interpreted within a cosmological geometry with three spatial dimensions. For the component considered we assume the balance $dU = -P\,dV$. We also assume that there are no additional exchanges with matter.

Let us read the formula in words. U is the energy contained in the region. V is its volume. P is pressure. The letter d indicates a very small change. The minus sign specifies how energy changes when volume changes in this balance.

If energy increases during expansion, the balance assigns a negative pressure. If energy is proportional to volume raised to λ, a simple result is obtained:

$$w = \frac{P}{\rho} = -\lambda. \qquad (4)$$

Here ρ denotes energy density, that is, energy divided by volume. The number w compares pressure with energy density. The two quantities have the same physical dimensions. In the example with $p = 1/4$, w is about -0.809.

The result has a precise meaning. A discrete rule selects a growth rate. With the step to physics just assumed, that rate determines an effective pressure. We have not chosen the value of w directly. We have, however, chosen the microscopic rule. We have also chosen the way of translating counts into physical quantities. Both are assumptions to be assessed.

Why a negative pressure can change the expansion

In everyday life we think of pressure as something that pushes against a wall. In cosmology we must not imagine it as a wind that carries galaxies away. It is a property of the component that enters the energy balance and the equations of gravity.

In general relativity, pressure also contributes to the rules of expansion. For a uniform component, the decisive term contains energy density plus three times pressure. If pressure is sufficiently negative, that contribution can support the acceleration of the *scale factor*. The scale factor is the number that indicates how cosmic distances grow. For a dominant component, the threshold is $w < -1/3$ [2].

Here we assume an effective theory of gravity. Counting records has not, by itself, proved Einstein's equations. The proposal derives a component to be inserted into those equations. It states the assumptions used.

Figure 4. Values of the regimes obtained from the rules, not measurements. The green curve uses the original branching rule. The blue curve halves the probabilities that produce new activity. It does so in the two channels described in Appendix C. In the dominant case both remain below the acceleration threshold.

The blue curve answers a natural question. Does the result disappear as soon as we touch a rule? In this family of models, no. We can reduce productivity and retain a range of accelerating behaviours. Appendix C indicates which productivity values guarantee this result for every redundancy probability considered.

It is not enough, however, for the distal component to cross the threshold on its own. The model's universe also contains matter, which contributes to the dynamics. The overall acceleration depends on the balance among all the components. The complete formula is in Appendix D.

How much space and how much time?

If the number of slots represents a three-dimensional volume, eight times as many slots correspond to a doubled length. Two cubed is eight. This leads to the *scale factor*, denoted by a. The factor describes how the distances change between references that follow the mean expansion.

Let us compare the epoch in which the light is emitted and the one in which we observe it. Between these two epochs, the assumed step between capacity and volume gives:

$$\frac{a_o}{a_e} = \left(\frac{n_o}{n_e}\right)^{1/3} = Q_{e,o}^{-1/3}. \qquad (5)$$

The subscripts indicate which epochs we are comparing: "e" for emission and "o" for observation. The cumulative link is thus connected to the increase in physical lengths. This happens because we have identified capacity with volume.

But a doubling can take different amounts of time. Knowing how much a quantity grows does not yet say how quickly it grows. Counting the updates does not automatically provide a clock.

Figure 5. Two different rates, but the same final growth. The overall product of the links is also the same. One curve grows linearly, the other accelerates. It is a kinematic example. It shows that capacity alone does not determine the rate. It does not present two solutions of the same complete physical dynamics.

To obtain the rate we adopt cosmological proper time. It is the time measured by ideal observers who follow the mean expansion. We also adopt the Friedmann equations. These equations connect the expansion rate with the density of the components and with gravity. Together with the energy balance, these rules close the uniform system. Given the initial conditions, we can calculate how the history unfolds.

The clock is therefore connected consistently to the effective dynamics. It has not been derived from the algebra of the link alone. This distinction lets us reach a physical prediction without crediting the network with a deduction we have not made.

What does "distal" mean?

When we look far away, we receive light that set out in the past. Observed distance and the history of the universe are therefore tied together in the same light. Here *distal* denotes the links that accumulate between two epochs being compared. An epoch is a moment in the history of the universe, like a date in a diary.

This does not mean that a region has a higher density today simply because it is far from us. Take another observer who follows the same mean expansion. That observer does not become the physical centre of a different universe. The reference point of the comparison changes. The cosmological law must remain the same. It is the shared rule for describing the expansion.

Light lets us check whether this connection is consistent. In the adopted geometry, wavelengths stretch together with the scale factor. A wavelength is, for example, the distance between two crests of a light wave. We call *redshift* the relative increase of that length. If the length doubles, the redshift is one, not two. The stretching factor is $1 + z$.

Two light crests emitted a short time apart also arrive farther apart in time. For very small intervals, the same relation holds:

$$1 + z = \frac{d\tau_o}{d\tau_e} = Q_{e,o}^{-1/3}. \qquad (6)$$

In words, the formula says that three things have the same factor. They are the redshift, the stretching of time intervals and the growth between the two epochs. Here e indicates emission and o observation. The letter d indicates a very small change. The symbol τ is cosmological proper time. It is the time followed by observers who take part in the mean expansion. The symbol Q denotes the share of correlation transmitted along the links. For finite intervals, this approximation holds if the intervals are short compared with cosmological changes.

This relation is not a new account of quantum behaviour. Nor does it derive local relativity from the binary channel of the registers.

[IMPORTED EFFECTIVE POSTULATE] Light and the meaning of clocks follow standard cosmological geometry. In the connection used to calculate luminosity distance, the number of photons is conserved. Photons are the particles of light.

This last qualification avoids counting the same effect twice. We do not multiply the luminosity by Q again. We do not treat this fraction as the probability that a photon survives. The fraction belongs to the relational register. Light is already described by the adopted propagation law [1].

What do we ask of supernovae?

A cosmological theory must confront measurements. It must not only propose suggestive images. Here we use an already completed comparison with Type Ia supernovae. They are stellar explosions. Their light curves, after some corrections, help to estimate distances. They are not all identical light bulbs. The analysis takes into account their standardization and the uncertainties [3].

The model supplies an expansion history. From that history we calculate the relation between redshift and apparent brightness. Apparent brightness indicates how bright the supernova looks from Earth. The question is restricted. A curve is fixed before the new calculation. Does it remain comparable with other observations, without changing its physical shape to fit the new points?

The first data set served to determine the parameters. It contained Pantheon+ and seven DESI BAO measurements, up to redshift one. BAO denotes a characteristic scale in the distribution of cosmic matter. This scale is used as a reference for measuring distances. This work is described in Section B18 of the MM 2.30 volume [11, 14].

The subsequent comparison used the DES-Dovekie reanalysis. A reanalysis is a new examination of data already collected. Objects already associated with the comparison inventory were excluded. Two uncertain associations were also excluded. Under the documented criteria, 1,448 distinct DES objects remain.

Pantheon+ also had 1,448 unique objects in the first comparison. It is a numerical coincidence between two different sets. It is not the same sample renamed.

The physical parameters stayed at the values used before. On the new observations only a uniform shift of the brightness scale was recalculated. This shift is needed when the absolute calibration is not fixed. Calibration is the reference that connects the measured numbers to a known brightness. Moving the whole curve up or down is not the same as changing its shape.

The procedure was fixed before acquiring and calculating the individual new distances. But it was fixed after the published DES results as a whole were known. **This was neither a blinded test nor an external preregistration.** A blinded test hides some results from the researchers until the end. An external preregistration records the procedure in advance with an independent party.

Even catalogues without shared objects might share the same calibrations. These conditions limit the strength of the comparison. They are part of the result and must be stated.

The curves are close to the data

How do we summarize a comparison with more than a thousand measurements? One of the quantities used is called *chi-square*. It summarizes the differences between curve and data. It takes into account the uncertainties and their

correlations. In this comparison, with the same procedure, a lower value indicates smaller overall differences.

Model applied to the DES data	Chi-square
Cosmological-constant reference	1263.358
Control model with constant w	1264.875
MM B17 distal component	1265.041

Table 1. Results from the earlier local study. They were not recalculated as a new test for this edition. Each model retains its own matter parameter, calibrated in B18. The control model with constant w also retains the vacuum residue. The residue is specified in Appendix G. The differences do not become probabilities that a theory is true.

The three curves are very close. MM is not preferred. Its chi-square exceeds that of the cosmological-constant reference by about 1.68. There is also the comparison with the control model with constant w. It shows that these data distinguish the distal mechanism only a little from another expansion history.

A restricted result remains valid. In this relative comparison, the model reaches a curve. This curve does not depart sharply from the adopted references. This does not show exclusively that relational links are the origin of the expansion.

There is a further statistical caution. The absolute chi-square values are low compared with a simple Gaussian reading. The DES procedure uses weights and corrections. So we cannot directly turn these values into a reliable percentage of theoretical validity.

A difference in the effective number of measurements reported in the publication remains to be clarified. Appendix H presents the numbers without hiding them. It does not give them a meaning they do not have.

For this reason we will not use "test passed" as a stamp of approval. We will say exactly what happened. The curve was transferred keeping the physical parameters fixed. The relative comparison is close to the control models. MM is not preferred. No absolute validation probability is available.

How far does this story reach?

The comparison with observations stops at redshift one. In terms of the scale factor, it follows a precise phase. It is the phase in which physical distances between references that take part in the expansion have grown up to twice their value at the chosen past endpoint. This solution does not reconstruct the entire history of the universe.

The boundary is not merely a choice of writing. Let us extend the solution kept fixed backwards. The mean number of type-B registers reaches zero near $z = 1.665$. Going further, it would become negative. A negative number of registers is not an admissible state. So that continuation must be stopped or replaced with another physical connection. Appendix H documents the check.

Within the analysed interval, the solution remains admissible. We cannot, however, use the same history, unchanged, for the primordial universe. In particular, we do not present an MM comparison with the cosmic microwave background. Nor do we present one with all the BAO measurements at high redshift.

The model nevertheless has consequences that can already be put to the test within its interval. With the stated rules and physical connection, the pressure-to-density ratio of the distal component does not fall below -1. In the curve kept fixed it is about -0.815 today. It is about -0.875 at redshift one.

The value of about -0.809 illustrated earlier belongs instead to the dominant regime of the dynamics. It is not the present value of the complete curve. A dominant regime is the part of the evolution in which the most persistent growth mode prevails.

Let us think of a robust and comparable observational reconstruction. If it required a behaviour incompatible with these bounds, it would challenge this version of the model. The DESI analyses discuss preferences for a varying dark energy. These preferences depend on which data are combined and on how the parameters are represented. They are a

comparison to be addressed. They are not an automatic refutation inferred from the headline of a result [9].

Being able to update a theory does not allow its rules to be changed after every result without recording the change. An extension beyond the current interval would be a new or larger model. It would have its own assumptions and checks. Here we present the restricted contribution that we have built.

A proposal within the scientific conversation

The idea that geometry might have a more fundamental description did not begin with MM. Recognizing neighbouring work helps to understand which question we share. It also helps to understand how our construction differs. This recognition does not assign to MM the results obtained by others.

In **causal sets**, spacetime is described with discrete elements. It is also described with an order that represents cause-and-effect relations. The number of elements is connected to the volume of spacetime [4]. There is a similarity with the step from counts to geometry. But in our model the spatial volume of a region at a given epoch is not their spacetime volume. Moreover, our binary channel does not already contain that causal order.

In **Everpresent Lambda**, Ahmed, Dodelson, Greene and Sorkin study a varying, fluctuating cosmological constant. It is inspired by unimodular gravity and causal sets [5]. Here we instead follow the growth and dilution of an active component. We keep a possible constant part of the vacuum separate. The fluctuations of our counts do not automatically inherit the predictions of that programme.

Padmanabhan proposed interpreting expansion through an imbalance. It is the imbalance between the degrees of freedom associated with a surface and those in the interior [6]. The connection lies in associating expansion with counts. The mechanism is different. Here MM does not calculate those two numbers and does not derive their balance law.

Oriti, Sindoni and Wilson-Ewing derive cosmological dynamics from particular collective states in group field theory. They also use a scalar field as a reference for time [7]. It is another route that goes from a microscopic description to cosmology. It uses quantum structures of its own. Our statistical model does not replace it through a simple analogy.

Finally, the mathematics of branching belongs to a family studied for a long time. These are the **generalized Pólya urns** and the processes with several types, including Janson's work [8]. One draws an element and changes the composition according to rules tied to its type. This formalism was not invented by MM.

The difference of this construction lies in its starting element. We do not choose an even smaller piece of geometry, as many of these programmes do. We choose a relation: the possibility of telling two states apart. Space does not contain these differences. It counts them.

The proposed contribution is the specific chain that connects four elements. They are the fractionating channel, capacity, active growth and the cosmological connection. To establish whether and to what extent this chain is original, a broader comparison with the literature is needed. We do not claim general priority for the idea that space emerges from degrees of freedom.

How can the network grow without stretching every structure?

The capacity model has brought us to an expansion history. Now the question is more precise: must every new register enlarge space? If it did, how could a structure keep its size while the network around it grows?

Let us imagine a survey of a changing town. Some new entries confirm a connection already described. Others indicate a new intermediate place and change the route between existing places. Both add information to the archive. Only the second changes the spatial description of the town. MM can express this difference as a candidate rule, instead of leaving it as a simple comparison.

[CANDIDATE MM RULE] We assign a symbol to every spatial register. We read the endpoints of a link through a noisy channel. If the readings are equal, a memory is created. If they are different, an interposition is requested. It is a new spatial register between the endpoints. The request is realized only if the neighbouring cells can be updated compatibly. Otherwise it remains a memory of an unrealized distinction.

Figure 6. Two kinds of record, not two kinds of substance. Checking compatibility may require further comparisons. Each has its own reading and its own outcome. The drawing shows a candidate update rule. It is not an observation of microscopic space.

This is a new exploratory model [16], not a new name for B17. It distinguishes spatial registers, memories and the count to which energy might be assigned. The supernova comparison in Table 1 still belongs only to the original B17 construction.

Now we can ask a more precise question. Which relational changes alter the current spatial organization? And which preserve a memory without doing so? Appendix I defines the rule. The following appendices show what can already be calculated with it.

How does a zone stay stable in a growing network?

Consider a region in which all registers have the same symbol. Its internal links transmit the signal without errors. Every internal reading then confirms agreement. Under the proposed rule, these comparisons add memories. But they do not divide the spatial links of the region.

Outside that region, imperfect links can produce different readings. Compatible interpositions increase the distances measured along the surrounding spatial network. The same update rule can therefore keep an organization and make the environment grow. There is no need for a separate command ordering that this particular object be held fixed.

The exploratory mesh model gives a numerical result on this mechanism. Let us consider the recorded simulation with a coherent region. Here the mean internal distance on the network stays about 1.32. This holds up to a total of 131,072 records. In a control, the links in the region are imperfect. Here the internal distance instead grows to about 6.96 [16]. These are steps in the network, not metres. The simulation is an experiment internal to the model, not an astronomical measurement.

What is given at the start matters. We supply the model with coherent states and perfect internal links. The simulation shows that this preparation can last. It does not explain how ordinary matter first forms organized in that way. The structure of the cells in three dimensions is also supplied. The dynamics does not select it from a network without structure.

The model distinguishes two descriptions that must not be confused. The cells can be subdivided inside a reference region. The coordinate volume of this region stays fixed. Meanwhile, the paths measured in steps of the current network can become longer. To interpret this second change as physical expansion, a physical calibration is needed. A denser network, by itself, does not prove that the universe has expanded.

Even keeping this limit in mind, we have obtained a useful result. Expansion need not mean the same behaviour for every relational organization. We can study stability and growth with a single explicit candidate mechanism. Formation, the calibration of physical lengths and gravitational binding remain steps to be built. They are not hidden ingredients of the demonstration.

How do we count the work needed for a comparison?

A new interposition may first require adjustments in the neighbouring links. Let us imagine moving a junction. The surrounding roads must also continue to connect consistently. In the mesh of cells, this involves a chain of further comparisons. A requested change may fail. The earlier successful changes nevertheless remain valid events. This is because each has its own reading and its own record.

Counting only the initial request is not enough. We need the mean numbers of spatial registers and memories produced by a complete cell event. The necessary comparisons are included in the event. We denote them by the growth coefficients γ_S and γ_{mem}. A growth coefficient says how much we expect a quantity to increase.

An idealized chain model makes it possible to calculate these averages. It assumes that the required requests are drawn independently. It also uses a fixed confirmation probability. Its closed equations remain useful. But a network retains information. Two links with the same number of records can have different states. So they can have different probabilities of creating a new spatial register.

The next step is now explicit. We keep the states, the current cells and the order of the compatibility checks. The reading rule then determines the probability of the possible histories of an event. We sum the spatial outputs and the memories of these histories to define the growth coefficients. Appendix O presents this formulation. The formulation depends on the state. There is no need to insert an average confirmation probability into the microscopic evolution.

Eight new simulations of homogeneous networks follow this rule. They reach completed refinement levels. With a

memoryless channel, the mean production per cell event is about 0.928 spatial registers and 0.432 memories. Let us then consider a channel that retains half of the symbol signal of each endpoint. With this channel the values become about 0.893 and 0.458. The measured confirmation share rises from about 0.249 to 0.266. These are tests on finite networks, not astronomical observations.

The difference has a precise meaning. Retaining information changes growth even when the rule for recording stays the same. Appendix P proves that the total numbers of spatial registers and memories alone are not enough to determine this evolution. Appendix Q describes the complete groups of runs and the independent random seeds. It also describes the limit due to finite size. The historical short-run comparisons remain identified as such in Appendix L.

The independent-chain formula still approximates the new results when it receives the measured statistics. It does not thereby become an autonomous prediction. A more recent approximation is based on pairs of links. It improves some geometric averages. But it fails a consistency check, documented in Appendix R. Neither of the two approximations is needed to execute the rule that follows the state.

The result reached is restricted but complete. It includes a specified candidate process and reproducible tests up to whole levels. It is not a unique microscopic theory derived from MM's ontology.

What changes if we choose energy differently?

We can now return to the connection between counts and physics. We use a more structured network. The original B17 construction assigns volume to total capacity. It assigns energy to the number of active registers. The exploratory network instead distinguishes spatial registers and memories. It therefore offers a different candidate connection. Volume is proportional to the number of spatial registers. Energy is instead assigned to a specified group of records.

If every spatial register has the same effective energy contribution and the same volume contribution, their ratio stays constant. With the separately adopted work balance, the pressure-to-density ratio is minus one. This is a possible microscopic interpretation of a constant-density component. It is not a new fit to the data. It does not replace the changing active fraction used by B17.

[IMPORTED EFFECTIVE POSTULATE] We assign physical volume and energy to the selected counts. For the component considered we use the work balance without additional exchanges. Appendix N gives the general relation between their growth rates. The counting rule, by itself, fixes neither the unit of energy nor that of volume.

It is also possible to assign energy to the unrealized distinctions. In that case their accumulation near a coherent region matters. The simulations show that the count of these distinctions increases strongly near the region, compared with a control. They do not, however, establish a physical energy profile or its gravitational field. The choice of energy must not be hidden inside the word "memory".

General relativity does not require energy to be assigned only to spatial registers. Its sources include energy, momentum and stresses. A register does not become energetically irrelevant merely because it creates no additional volume [15]. Assigning energy only to spatial registers is a minimal modelling choice within MM. It is not a consequence of Einstein's equations.

This extension sharpens the problem. But it does not change the observational result already reported. It distinguishes memories and spatial organization. It proposes a persistence mechanism. It makes it possible to calculate an approximation of growth. It also brings to light the physical decision that remains. Which distinctions carry energy, and with what coupling and calibration?

The answer may change as the microscopic model improves. In this edition, the cosmological equations of B17, the fitted parameters and the supernova comparisons remain unchanged. The new mechanism is a research extension alongside the effective construction already compared with observations.

From links to an expansion history

We started from the question whether relations could be more fundamental than objects and space. Now we can reconstruct and discuss the operations used in the construction.

The construction connects channels defined by integer counts, cumulative fractionation, growing capacity and active comparisons. A stated physical connection then carries the calculable growth to density, pressure, expansion and light signals.

We can indicate where each consequence comes from. We can change an assumption and calculate how much the result depends on that choice.

The value of the proposal does not require observations that are different or new compared with those used by other theories. Every description must confront observations. An alternative theoretical organization can be interesting if it makes explicit connections that previously remained separate. But agreeing with an already known trend does not, by itself, identify the physical origin of that trend.

Let us consider the extension that distinguishes memories and space. It adds the difference between recording and interposing. It adds the persistence of a prepared coherent region. It also adds conditional growth equations. It specifies the reading process that depends on the state. It proves why counts alone do not describe it exactly for all states. Moreover, it examines eight groups of runs with completed levels.

These results do not select B17's dynamics and do not inherit its observational comparison as if it were their own validation.

Open questions remain. What is the local origin of the collective rule? How do coherent regions form? How can space and clocks be derived from relational premises alone? How are energy and volume calibrated? How are fluctuations connected to observable perturbations? The value of the vacuum residue remains unexplained. These are limits of the construction presented, not promises added to the result.

The restricted thesis is this: an explicit relational model exists. Through a stated physical connection, this model generates a component capable of supporting an accelerated expansion. Moreover, it reaches a limited observational comparison. The comparison does not favour it over the alternatives. The construction remains a conditional mathematical and physical proposal, which can be assessed.

For those who want to check the steps

Appendices A-F retain the original derivations. Appendices G-H specify the data and the interval to which the result applies. Appendices I-N develop the alternatives on memory, space, chains and energy. Appendices O-R present the rule that depends on the state. They show why counts alone are not enough to close it exactly. They report the tests with completed levels. They also check the approximation based on pairs.

Appendices S-V contain the comparison with supernovae, baryon acoustic oscillations and the summary data of the cosmic microwave background. They also report the bounds and the point beyond which the treatment of the local noise is no longer valid. Finally, they report the test fixed before the calculation for the non-uniform sector. Both reading paths can be followed without knowing the conversation that came before them.

Readers who do not wish to continue can stop here. Those who want to go directly to the new observational result can go to Appendix S. Appendices A-R contain the derivations and the history of the model's development.

Appendix A. The link has a precise probability

[CANDIDATE MM RULE] A channel connects two registers, denoted by σ and σ'. Each can have state -1 or $+1$. For each pair of equal signs we count a internal possibilities. For each pair of different signs we count b. All possibilities have the same probability. The integers satisfy $a > b \geq 1$.

The matrix T collects these counts. I leaves each sign in its position. J swaps the two positions. Dividing by the total $a + b$ we obtain a probability matrix:

$$T = a\mathsf{I} + b\mathsf{J}, \quad P(q) = \frac{T}{a+b}, \quad q = \frac{a-b}{a+b}. \qquad (A1)$$

In words: the channel favours agreement over disagreement. The number q measures this difference after dividing it by the total of the possibilities.

We adopt an initial distribution in which the two signs are equally likely. This distribution stays the same after passing through the channel. Moreover, knowing the initial sign σ, the mean value of the final sign is $q\sigma$. The mean value summarizes many cases. It is like averaging many tosses of a coin. One assigns $+1$ to heads and -1 to tails.

Repeating the step along an open chain, we obtain a mean correlation between the endpoints. It is the product of the values q of all the links: $Q = \prod q_e$. The matrices share the same two eigenvectors. They are two special directions that do not change direction when the matrix acts on them. For this reason $P(q_1)P(q_2) = P(q_1 q_2)$.

On a tree, there is only one path between two endpoints. In a network with cycles, instead, there can be several paths. The product formula does not hold in general for the correlation of the joint distribution of that network.

[DEDUCTION WITHIN THE MODEL] If the path is unique, we can define $d_l = -\log Q$. It is a length without units of measurement. It is positive if every link has $q < 1$. It is not a Lorentzian metric, that is, the measure of space and time used in relativity. It does not determine how many dimensions space has.

Suppose that the relative capacity is a continuous and positive function $F(Q)$. We require that the product of two correlations multiplies the capacities: $F(xy) = F(x)F(y)$. We also require that capacity grows when Q decreases. We then define $g(s) = \log F(e^{-s})$. The function g is continuous. It turns sums of values of s into sums of values of g. These conditions imply $g(s) = \alpha s$, with $\alpha > 0$ if the function is not constant. Thus $F(Q) = Q^{-\alpha}$.

This form follows from the assumptions on the function. It does not follow from them that this capacity must be the number of slots, nor that α must equal one.

The version adopted here chooses $\alpha = 1$, $a_n = 2n + 1$ and $b_n = 1$. Multiplying the steps and adding the logarithmic lengths we obtain:

$$\prod_{n=n_i}^{n_f-1} P(q_n) = P\left(\tfrac{n_i}{n_f}\right), \qquad \sum_{n=n_i}^{n_f-1} (-\log q_n) = \log \tfrac{n_f}{n_i}. \qquad (A2)$$

The first formula concerns the product of the links between the two states. This product is equivalent to a single channel with value n_i/n_f. The second formula concerns the logarithmic lengths of the steps. They add up to the logarithm of the growth of capacity.

The case $\alpha = 1$ is a bridge assumption. It was not measured with an experiment and it is not a theorem about MM's ontology. The sequence of channels connects the epochs of the process considered. It is not a rule for every spatial branch.

The internal counts of the channel are neither the number of slots n nor the number of active registers K. Nor is the multiplicity descriptor W, introduced earlier, identified here with any of these three quantities.

Let us divide a link into several segments, while their product stays the same. In this case the relation described does not change. An update $n \mapsto n + 1$, instead, changes the product. Growth therefore does not depend on the arbitrary number of segments with which we draw the chain.

Appendix B. A single rule for capacity and registers

The state is (A, B, I, σ). The quantities A, B and I are nonnegative integer counts. Moreover, $A + B + I = n \geq 2$. At each step we choose a slot at random. We replace it according to the rules already described.

We denote by $\mathcal{K}_n(x, x')$ the transition rule for the counts. Here $x = (A, B, I)$ is the initial state and x' the next one. The probabilities of all possible outcomes sum to one. Every outcome increases the total of slots from n to $n + 1$.

[CANDIDATE MM RULE] Given the state and the value of n, we assume that the binary history channel and branching are independent. The joint probability becomes:

$$\Pr(x', \sigma' \mid x, \sigma) = \mathcal{K}_n(x, x') P(q_n)_{\sigma\sigma'}. \qquad (B1)$$

The formula is normalized. The probabilities are nonnegative and sum to one. It realizes at the same time the multiplication of channels of A2 and the dynamics of the active registers. It does not say that one of the two rules automatically imposes the other. The law depends on global capacity. A local causal mechanism that generates it remains to be derived.

Let us now consider $X = (A, B)^T$. It is the column that collects the counts of the two active types. The possible changes are $v_1 = (0, 1)^T$, $v_2 = (1, -1)^T$ or no change. Their probabilities are, respectively, $[A + (1 - p)B]/n$, pB/n and I/n. The mean of the change is:

$$\mathbb{E}[\Delta X \mid X] = \frac{MX}{n}, \quad M(A, B)^T = (pB, A + (1 - 2p)B)^T. \qquad (B2)$$

The matrix M collects how much each type contributes to the mean change of the next step. We call $m_n = \mathbb{E}X_n$ the mean of the counts after step n. We obtain exactly:

$$m_{n+1} = (\mathrm{I} + \frac{M}{n})m_n. \qquad (B3)$$

It is exact because the mean increase depends linearly on the current means and the total n is determined, not random.

Let us take an eigenvalue λ of M. It is a number that describes one of the matrix's own growth rates. The accumulated factor between n_i and n_f is:

$$\prod_{n=n_i}^{n_f-1} (1 + \frac{\lambda}{n}) = \frac{\Gamma(n_f + \lambda)\Gamma(n_i)}{\Gamma(n_f)\Gamma(n_i + \lambda)}. \qquad (B4)$$

The function Γ extends the factorial to non-integer values as well. If n_i becomes large and we keep the ratio n_f/n_i fixed, the formula approaches $(n_f/n_i)^\lambda$. The relative error is of the order of $1/n_i$.

This limit leads to the continuous equation $dm/du = Mm$, where $u = \log(n/n_i)$. It is a controlled approximation for the means. To approximate a single random history as well, the fluctuation control of Appendix F is needed.

Appendix C. Why active growth is robust

The eigenvalues of M in B2 are those of Formula 3. Both signs in front of the square root are taken. For $0 < p \leq 1$ and nonzero initial activity, the dominant mode is accessible and positive. The dominant mode is the growth direction that prevails in the long run.

In the continuous limit, we call $\bar{K} = m_A + m_B$ the mean active total. We call $f = m_B/\bar{K}$ the mean share of type B. We obtain:

$$\frac{d\log\bar{K}}{du} = 1 - pf, \qquad \frac{df}{du} = 1 - (1 + 2p)f + pf^2. \qquad (C1)$$

Here f is the ratio between two means. It is not the mean of the random ratio B divided by A+B.

The value of f stays between zero and one. If it is zero, its rate of change is one. If it is one, the rate is $-p$. So the rule does not push it out of the interval. Moreover, let us take two solutions f and g with the same p. The difference between their fields of change is $-(f - g)[1 + p(2 - f - g)]$. The distance between the solutions therefore cannot be greater than $e^{-u}|f(0) - g(0)|$. The different initial conditions come closer at least according to this bound.

The stable fixed point is $f_* = (1 + 2p - \sqrt{1 + 4p^2})/(2p)$. At that point the growth rate is $1 - pf_* = \lambda_+$. The approach to this value depends on the same initial conditions. This transient phase is not removed before the comparison with observations.

Let us also examine a broader family of rules, with reduced productivity. A produces A+B with probability s. Otherwise it produces A+I. B produces A+I with probability p, B+B with probability $(1-p)t$ and B+I in the other cases. All probabilities remain admissible when $s, t \in [0, 1]$.

The matrix of the mean contributions in this family is:

$$M_{s,t}(A, B)^T = (pB, sA + bB)^T, \quad b = t(1-p) - p. \qquad (C2)$$

The dominant rate is:

$$\lambda_+ = \frac{b + \sqrt{b^2 + 4sp}}{2}, \quad \lambda_+ > \frac{1}{3} \iff t + p(3s - t - 1) > \frac{1}{3}. \qquad (C3)$$

The other root is not positive. To verify the condition in C3, one evaluates the characteristic polynomial $x^2 - bx - sp$ at the point $x = 1/3$. The inequality has a strictly positive margin for every $p \in [0, 1]$ exactly when $s > 4/9$ and $t > 1/3$. It is a result about the model, not a measurement of parameters of the universe.

If $p = 0$, the redundant channel disappears. In this case the original rule gives $\lambda_+ = 1$. It is a formal limit. With a finite number of labels having normalized probabilities one cannot have $\sum \pi_j^2 = 0$. Moreover, that limit does not automatically eliminate the fluctuations.

Appendix D. The physical bridge and the complete clock

[IMPORTED EFFECTIVE POSTULATE] Consider a homogeneous region that follows the mean expansion. It is called comoving because it keeps its identity while its physical volume changes. We set $V = v_* n$ and $U_D = E_* \tilde{K}$, with positive constants. V is the volume and U_D the energy of the distal component. The units assigned to each count are v_* and E_*.

In three dimensions, the scale factor satisfies $a/a_i = (n/n_i)^{1/3}$. We define the energy density $\rho_D = U_D/V$. We assume the pressure $P_D = -dU_D/dV$, without additional exchanges with matter. From these assumptions it follows:

$$w_D = -\frac{d\log \tilde{K}}{du} = -1 + pf, \qquad \frac{d\rho_D}{du} = -(1 + w_D)\rho_D. \qquad (D1)$$

The first formula concerns the ratio w_D between pressure and density. This ratio depends on the growth rate of the active registers. The second describes how the density changes. In the general case of C2, $w_D = -s(1-f) - t(1-p)f$. Since the values of s, t, p, f lie between zero and one, $-1 \le w_D \le 0$.

In the dominant mode, the density follows $\rho_D \propto a^{-3(1-\lambda_+)}$ and $w_D = -\lambda_+$. These results depend on the effective connection assumed between counts and physics.

[IMPORTED EFFECTIVE POSTULATE] We adopt cosmological proper time τ. It is measured by ideal observers who follow the mean expansion. We assume a flat FLRW geometry and general relativity. We also assume separately conserved nonrelativistic matter and a possible constant component ρ_v. FLRW denotes the homogeneous and isotropic model used to describe the expansion. We neglect radiation in the late period considered here.

Keeping the speed of light c explicit, the closed system is:

$$H^2 = \frac{8\pi G}{3c^2}(\rho_m + \rho_D + \rho_v), \quad \frac{du}{d\tau} = 3H, \quad \frac{dm}{d\tau} = 3HMm. \qquad (D2)$$

H is the expansion rate, G the gravitational constant, and ρ_m the matter density. We choose the branch $H > 0$, a positive total density and the initial conditions. With these choices we can calculate the evolution. Matter follows $\rho_m = \rho_{m,i} e^{-u}$.

Differentiating the Friedmann constraint and using D1 and D2, we find:

$$\dot{H} = -\frac{4\pi G}{c^2}[\rho_m + (1 + w_D)\rho_D]. \qquad (D3)$$

The derivative with the dot is with respect to proper time τ. The acceleration of the scale factor is also:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3c^2}[\rho_m + (1 + 3w_D)\rho_D - 2\rho_v]. \qquad (D4)$$

The formulas make it possible to check that the constraint, the energy balance and the acceleration are consistent. No growth law incompatible with the energy balance is added.

If only the distal component dominates and $\lambda_+ < 1$, the scale factor follows $a \propto (\tau - \tau_*)^{2/[3(1-\lambda_+)]}$. If instead $\lambda_+ = 1$, it follows $a \propto e^{H\tau}$. These are solutions valid in those regimes. They do not reconstruct the Big Bang or the actual age of the universe. When matter is present, D4 must be used. Checking only that $w_D < -1/3$ is not enough.

Appendix E. From the universe's rate to light signals

[IMPORTED EFFECTIVE POSTULATE] We assume that light follows the null geodesics of the FLRW geometry. We also assume that the number of photons is conserved. A null geodesic is the path that light follows in this geometry. For a comoving emitter and receiver, the radial path satisfies:

$$\chi = \int_{\tau_e}^{\tau_o} \frac{c\,d\tau}{a(\tau)}.$$

The quantity χ represents the comoving distance. It is the distance measured taking the expansion into account. Two light crests emitted a short time apart travel the same comoving interval. Subtracting the two integrals we obtain:

$$\frac{d\tau_o}{a_o} = \frac{d\tau_e}{a_e}, \qquad 1 + z = \frac{a_o}{a_e} = Q_{e,o}^{-1/3}. \qquad (E1)$$

The formula connects cosmological events. It does not derive from the algebra of the registers the local time dilation predicted by relativity.

We set today's $a_o = 1$. We denote by $\mathcal{E}(z) = H(z)/H_0$ the expansion rate divided by its present value. In the flat case, the luminosity distance is:

$$D_L(z) = (1 + z)\frac{c}{H_0}\int_0^z \frac{d\zeta}{\mathcal{E}(\zeta)}. \qquad (E2)$$

To compare the theory with supernovae we use the distance modulus, $\mu = 5\log_{10}[D_L/(10\,\mathrm{pc})]$. We use the usual calibrations and an additional term if the absolute luminosity is not fixed. The parsec, denoted by pc, is a unit of astronomical distance. E1 and E2 are the standard cosmological bridge [1]. They are not a new law of light propagation.

To calculate $\mathcal{E}$ without using only the limit of the dominant mode, we set $a_i = 1/2$, $p = 1/4$ and $f_i = 1/2$. The initial normalization of m_i cancels in the ratio:

$$D(a) = \left(\frac{1}{a}\right)^3 \frac{\mathbf{1}^T e^{3M\log(a/a_i)} m_i}{\mathbf{1}^T e^{3M\log(1/a_i)} m_i}. \qquad (E3)$$

Here $\mathbf{1}^T = (1, 1)$ adds the two active types. The matrix exponential function applies the continuous evolution defined by M. The total normalized density of the model is:

$$\mathcal{E}^2(a) = \Omega_m a^{-3} + \Omega_v + (1 - \Omega_m - \Omega_v)D(a). \qquad (E4)$$

The function D equals one today. For B17, the inherited comparison uses $\Omega_v = 0.02$ and $\Omega_m = 0.2767981374$.

The cosmological-constant reference uses $\Omega_m = 0.3272014632$. These parameters and the normalizations come from the local study, not from the links alone. The transferred comparison holds for $0 \le z \le 1$.

The new interpretation of A2 does not change E3 and E4. It completes an interpretive connection compatible with the same capacity history. This article does not claim for A2 an improved fit, a new data selection or an observational validation. In the inherited comparison an unknown correlation between calibration and check is assumed to be zero. It is not proved that it is really absent.

Appendix F. Fluctuations, calibrations and alternatives

The discrete model lets us follow the fluctuations before giving them an astronomical meaning. We denote by $\Sigma_n = \mathrm{Cov}(X_n)$ the covariance matrix of X_n. A covariance describes how much several quantities vary together. It is like a table that summarizes the differences and the connections between measurements.

For the original rule, we define $D(m) = [m_A + (1-p)m_B]v_1 v_1^T + p m_B v_2 v_2^T$. The conditional second moment of the changes is $D(X)/n$. Applying the balance of the second moments, we obtain exactly:

$$\Sigma_{n+1} = \Sigma_n + \frac{M\Sigma_n + \Sigma_n M^T + D(m_n)}{n} - \frac{(Mm_n)(Mm_n)^T}{n^2}. \qquad (F1)$$

Suppose that the initial state is fixed without randomness. Suppose also that the ratio n/n_i stays below a fixed bound R. The possible changes are bounded. Formula F1 then gives an estimate $S_{n+1} \le (1 + C/n)S_n + C'$. Here $S_n = \mathrm{tr}\Sigma_n$ and the constants depend on R and on the rule. Repeating the calculation, S_n is of the order of n_i. It follows that $\mathrm{Var}(K_n/n)$ is of the order of $1/n_i$.

Chebyshev's inequality thus makes it possible to justify the limit of the ratios on finite intervals of relative capacity. It does not justify a uniform limit for all future times when n_i stays fixed.

To translate F1 into a spatial spectrum, into gravitational lensing effects or into anisotropies, three ingredients are needed. They are correlations between regions, causal transport and a rule that reduces the details at large scale. **This observational signature has not yet been derived.** Separating the counts does not determine the physical sound speed. We cannot simply assume that the parameter w equals the square of that speed.

The importance of the physical connection becomes clear if we change it. Suppose that $V \propto n^\nu$ and that the energy of each register grows as n^η. In the dominant mode, one obtains $w = -(\lambda_+ + \eta)/\nu$. The prediction therefore also depends on how we connect the counts to physical quantities. It does not depend only on the matrix. Adding $U_v = \rho_v V$ leaves the counts unchanged. But it changes the cosmology. The vacuum residue remains distinct and unexplained.

Finally, a homogeneous history with $\rho_D > 0$ and $-1 < w_D < 1$ can be reproduced by a canonical scalar field. A scalar field is a quantity that has a value at every point of space. Setting $c = 1$, the relations are $\dot{\phi}^2 = (1 + w_D)\rho_D$ and $V_\phi = (1 - w_D)\rho_D/2$ [2].

This is an alternative effective description. It does not derive the microscopic field from MM. It shows why agreement on distances does not uniquely identify a relational origin.

Appendix G. How to read and reconstruct the comparison

Table 1 takes up a local transfer calculation [14]. It is not a new fit. There is also a separate and wider comparison, which fits the parameters again. It is reported in Appendices S-U and does not modify that table.

The parameters come from B18: 1,565 Pantheon+ light curves, 1,448 distinct objects and seven BAO measurements within redshift one.

Model	Matter parameter kept fixed

B17	0.2767981374
Cosmological-constant reference	0.3272014632
Control with constant w	0.2713974385

The control model retains $w = -0.8151435616$ for the nonconstant component. It also retains $\Omega_v = 0.02$, as in B17. So it is not correct to describe it, without qualification, as a model with a single dark component having constant w. In the cosmological-constant reference, the entire component that is not matter is constant.

The DES selection retains survey code 10 and the interval $0.10 \le z_{HD} \le 1$. In this interval there are 1,602 DES objects. Of these, 152 matching the Pantheon+ inventory are excluded. Two uncertain associations are also excluded. 1,448 objects remain, distinct from the calibration sample.

The number of objects coincides with that of the original Pantheon+ sample. But the samples are different. Having distinct objects does not guarantee that the standardization and calibration procedures are independent. An unknown correlation between the samples is assumed to be zero. It is not proved that it is absent.

The local procedure was fixed before acquiring the individual data and calculating the new probability of the data. The published DES results as a whole were already known. **It was neither a blinded test nor an external preregistration.** A blinded test hides some results from the researchers until the end. An external preregistration records the procedure with an independent party.

Following the DES convention, the shape of the distance modulus to compare is:

$$\widetilde{\mu} = 5\log_{10}\left[(1+z_{HEL})\int_0^{z_{HD}} \frac{d\zeta}{\mathcal{E}(\zeta)}\right]. \qquad (G1)$$

The two redshifts use different corrections for the reference frame. The additive constant takes the absolute calibration into account.

We denote by $r = y - \widetilde{\mu}$ the residuals, that is, the differences between data and curve. We denote by C the covariance matrix. The matrix also takes into account the connections between the uncertainties. In the main analysis only one parameter is fitted, a common shift of the brightness:

$$\widehat{b} = \frac{\mathbf{1}^T C^{-1} r}{\mathbf{1}^T C^{-1} \mathbf{1}}, \quad \chi^2 = (r - \widehat{b}\mathbf{1})^T C^{-1} (r - \widehat{b}\mathbf{1}). \qquad (G2)$$

The NPZ files contain the precision matrix. It is the inverse of the covariance. To obtain the covariance used in the selection, one first inverts the full matrix. Then one selects the part needed. One does not directly select the corresponding part of the precision matrix. Nor does one add the error diagonal a second time. The source of the data and of the procedure is given in [10].

Appendix H. Statistical checks and the limit of the solution

Why we give no validation percentage

The compressed DES probability uses weights related in part to the classification of the supernovae. In the local study, a Gaussian approximation also propagates the uncertainty of the calibrated matter parameter. In this approximation, the ratio $1265.0414/1447$ is about 0.87425. The lower-tail probability is about 0.0002094.

The two control models also give values of the same order. They are diagnostics calculated with the Gaussian approximation. They are not calibrated p-values for the full DES procedure. Nor are they probabilities that MM is true or false.

The sum of the Ia probabilities in the complete data used in the study is 1714.22219. In the selected sample it is 1343.86677. The effective number 1,684 reported in the DES paper cannot be obtained from these sums. It is not yet clear which convention or provenance accounts for the difference. We do not replace it with a convenient choice of

degrees of freedom.

There is a separate check on the full sample. It uses the standard model and the official term that considers all the possibilities of the common shift. It reproduces the published value 1640.3 to the precision shown. This supports the correctness of the implementation of the probability of the data. It does not calibrate the absolute significance of B17. And none of these checks extends B17 beyond its interval.

Where the backward continuation stops

For $p = 1/4$, $a_i = 1/2$ and $m_i = (1/2, 1/2)^T$, we set $u = 3\log(a/a_i)$. The continuous solution for the mean number of B registers is:

$$m_B(u) = \lambda_+ e^{\lambda_+ u} + \lambda_- e^{\lambda_- u}, \quad \lambda_\pm = \tfrac{1 \pm \sqrt{5}}{4}. \qquad (H1)$$

Following the solution towards the past, the first zero is at:

$$u_* = \tfrac{\log(-\lambda_-/\lambda_+)}{\lambda_+ - \lambda_-}, \quad a_* = \tfrac{1}{2} e^{u_*/3}, \quad z_* = a_*^{-1} - 1. \qquad (H2)$$

Numerically, $z_* \simeq 1.66468666$. If we continue the same formula even further back, m_B becomes negative. A negative number of registers is not a state allowed by the counting model. For this reason the comparison remains restricted to $z \le 1$, where the solution is admissible.

In the curve kept fixed, D1 gives $w_D(1) \simeq -0.81514356$ and $w_D(a_i) = -0.875$. The example that considers only the dominant mode instead gives $w = -\lambda_+ \simeq -0.80901699$. They are different values and not interchangeable.

Changing the initial state or the extension into the past would require two things. A new version of the model would have to be declared. The relevant comparisons would also have to be repeated.

Appendix I. A candidate rule for memory and space

This appendix defines an exploratory model. It is distinct from the three-type B17 process, described in Appendices B-F. We denote by S the spatial registers and by M_{mem} the memories. We denote by $N = S + M_{\mathrm{mem}}$ their total.

Every spatial register carries a fixed symbol x_i. The symbol is chosen from a set of m symbols. The candidate rule excludes memories from the graph of spatial adjacencies. A graph is a mathematical drawing of points and links. Here it serves to calculate which registers are neighbours.

Copying a memory is an operation defined by the model. It is not a derivation of the existence of an observer. Nor is it an account of quantum measurement.

On a link e we use the reading channel r_e. In the exploratory notes this quantity is called q. It is not the epoch channel q_n of Equation 1.

At each of the two endpoints, the reading copies the stored symbol with probability r_e. Otherwise it picks one of the m symbols at random. All symbols have equal probability. We set $A_r = (1 - r^2)/m + r^2$ and $B_r = (1 - r^2)/m$. The confirmation probability is:

$$A_r = \tfrac{1 - r^2}{m} + r^2, \quad B_r = \tfrac{1 - r^2}{m}, \quad \Pr(\text{confirm} \mid x_i, x_j) = B_r + r^2 \mathbf{1}_{x_i = x_j}. \qquad (I1)$$

The symbol $\mathbf{1}_{x_i = x_j}$ equals one if the symbols of the two endpoints are equal. Otherwise it equals zero. The formula says that the probability depends on the quality of the reading. It also depends on the agreement between the initial symbols.

[CANDIDATE MM RULE] A confirmation creates a memory. Two different readings instead request the interposition of a spatial register between the endpoints. Every comparison needed to prepare the interposition also follows the same

reading rule.

If the request cannot be completed, a memory of an unrealized distinction is created. If it succeeds, a spatial register is created with a new symbol chosen at random. The interpositions that succeeded in the earlier steps remain. They remain even if a later request fails. Every reading thus has its own recorded outcome. The entire chain is not a single indivisible update.

The numerical implementation uses ordered subdivisions of tetrahedra. It checks that neighbouring cells can be updated compatibly. A tetrahedron is a solid figure with four faces. The model already assumes the three-dimensional cells and their order. It also already assumes the selection weights and the reading channel. It does not derive them from a deeper rule.

Spatial distance is the length of the shortest path in the current spatial graph. It is not the negative logarithm of a historical correlation. Here neither of the two measures is identified with a relativistic metric.

Consider a region initialized with all symbols equal and with $r_e = 1$ on the internal links. In that region every internal reading confirms. The rule protects those links from subdivision. The reported stability of the internal distances is a numerical result. It holds for this preparation of the network. It is not a general theorem excluding every change of the shortest paths caused from outside.

In the limit $r_e = 0$, the confirmation probability is $1/m$. It is so whatever the initial states and the previous readings. This recovers B17's independent coincidence probability. It does not recover its entire three-type transition rule. It does not recover the energy assignment or the cosmological predictions.

Appendix J. How we count the outcomes of an event

We denote by k the number of top-level attempts. In each state, we choose a cell event with probability $s = S/N$. Otherwise we copy a memory and add one memory.

Consider a cell event and the whole state at that moment. We define γ_S and γ_{mem} as the mean number of new spatial registers and new memories produced. We also count all the comparisons needed before the event is completed. These means can depend on the state. They need not be constants.

Let R_{done} be the number of realized readings and F the number of confirmations. Let D_{stop} be the number of unrealized distinctions. Let L be the number of memory copies. In the numerical model considered, these identities hold:

$$S - S_0 = R_{\mathrm{done}}, \qquad M_{\mathrm{mem}} - M_{\mathrm{mem},0} = F + D_{\mathrm{stop}} + L. \qquad (J1)$$

The checks of the outcomes verify J1 in the runs examined. A cell event can produce several records. It is not correct to suppose that every top-level attempt increases N by exactly one.

The mean changes, given the state, are:

$$\mathbb{E}[\Delta S] = s\gamma_S, \qquad \mathbb{E}[\Delta N] = 1 + s(\gamma_S + \gamma_{\mathrm{mem}} - 1). \qquad (J2)$$

The first formula says that spatial increases happen only in cell events. The second also includes the memory copies. The formulas are obtained by separating cell events from copies.

From these means we can build a deterministic approximation. It is a curve that describes the mean behaviour without following every random case:

$$\frac{ds}{dN} = \frac{1}{N}\left[\frac{s\gamma_S}{1 + s(\gamma_S + \gamma_{\mathrm{mem}} - 1)} - s\right]. \qquad (J3)$$

J3 is not an exact equation for the mean of a random ratio. To treat it as a convergence theorem, an adequate continuous limit, a control of the moments and sufficient regularity are needed.

Suppose that the coefficients stay constant and that the regime is dominated by memories, with $s \to 0$. In this case the

approximation gives $d\log S/d\log N \to \gamma_S$. This conclusion is conditional. Coefficients measured during a transient phase do not by themselves prove an exponent valid for very long times.

In the reference control, at $N = 131072$, $S_0 = 64$ and $S = 27132$. The 27,068 new spatial registers split into 17,241 realizations of the initial request and 9,827 realizations of prerequisites. The 103,940 memories split into 11,517 confirmations, 4,421 unrealized distinctions and 88,002 copies [16]. These are counts of the model, not observations.

Coefficients measured in the same run can be used. This makes it possible to reconstruct the growth only approximately. The maximum recorded discrepancy for S/N is 0.03042. Appendix K presents a distinct calculation of the coefficients. The calculation is based on an idealized chain rule.

Appendix K. Closed equations for independent chains

Let J be the number of requests needed to complete a node that distinguishes the endpoints, if all the prerequisites succeed. We use a distribution π^{top} for the initial requests. We use a distribution π^{f} for the forced ones. In the chain notes, J is called K. It is not an energy count.

The model assumes that the child subtrees are independent. It assumes that J is independent of the reading outcomes. It also assumes that the confirmation probability is constant. We call this probability p and set $\alpha = 1 - p$.

The generating functions collect the probabilities of the different values of J. The first is:

$$G_f(R) = \sum_{j \ge 0} \pi_j^f R^j, \qquad H_f(R) = \sum_{j \ge 0} \pi_j^f \sum_{h=0}^{j-1} R^h. \qquad (K1)$$

The inner sum is zero when $j = 0$. We define G_{top} and H_{top} in the same way.

Let R be the probability that a forced request succeeds. To succeed, its reading must distinguish the endpoints and all the children must succeed. Hence $R = \alpha G_f(R)$. We choose the fixed point obtained recursively by increasing the depth of the chain. Here we use the subcritical case. It is the case in which the expected growth of the subtrees stays below the threshold that would make the chain explode. In this case the fixed point is unique.

Let T_f be the mean number of readings attempted in a forced chain. Let M_f be the mean number of spatial registers realized. The chain stops at the first child that fails. The earlier successful children, however, remain. It follows:

$$R = \alpha G_f(R), \qquad T_f = \frac{1}{1 - \alpha H_f(R)}, \qquad M_f = R T_f. \qquad (K2)$$

For an initial event, the means of spatial registers and memories are:

$$\gamma_S = \alpha[G_{\mathrm{top}}(R) + H_{\mathrm{top}}(R) M_f], \quad \gamma_{\mathrm{mem}} = 1 + \alpha H_{\mathrm{top}}(R) T_f - \gamma_S. \qquad (K3)$$

These are exact means for the independent-chain model. For them to be finite, the mean size of the subtrees must also be finite. A sufficient condition is $\alpha \mathbb{E}[J_f] < 1$. The distribution used here satisfies this condition.

The formulas do not assert that the events in the spatial mesh are independent. Applying them to that mesh is an approximation to be assessed. No metric, gravitational constant or cosmic clock appears in K1-K3. The formulas connect conditionally the local statistics of the prerequisites to the mean production of spatial registers and memories.

Appendix L. Proof and historical short-run comparison

Let us fix J. A child with index h is reached only if the preceding $h - 1$ children have succeeded. The probability of reaching it is R^{h-1}. Summing these contributions and averaging them over J we obtain H.

Independence lets each child reached contribute its own mean number of readings. The same holds for the mean number of spatial registers. So $T_f = 1 + \alpha H_f T_f$ and $M_f = R + \alpha H_f M_f$. These two relations prove K2. Repeating the same reasoning for the initial request we obtain K3. Every reading becomes either a spatial register or a memory.

The historical sample is based only on geometry. It contains 4,965 initial requests and 3,163 forced requests. It ends with 8,192 spatial registers. The mean number of required children is 0.533736 for the initial requests. For the forced ones it is 0.162188. These measured distributions are given in the English evidence materials [16].

With $p = 1/4$, the equations give $R = 0.717896$, $\gamma_S = 0.939370$ and $\gamma_{\mathrm{mem}} = 0.465357$.

Check	Spatial registers / memories produced on average
Closed equations of the independent chain	0.939370 / 0.465357
Separate simulation of 150,000 chains	0.940153 / 0.464367
Monte Carlo standard errors	0.002351 / 0.001839
Recorded spatial-mesh control	0.941234 / 0.554211

The Monte Carlo standard errors indicate how much the random error of the simulation could vary when the runs are repeated. The separate simulation checks the idealized equations, not the complete mesh.

The memory deficit of about 16 per cent belongs to this short comparison. It is not an estimate of the error for long times. The agreement on the spatial registers does not prove a universal accuracy of 0.2 per cent. Appendix Q tests a specific reading rule on completed levels. Samples and conditions are different. So its numbers do not replace those of this table.

The required number of children must be distinguished from the number actually visited. Imagine that two children are always required. Each succeeds with probability one half. On average we visit 1.5 of them, because the second is visited only when the first succeeds.

If we inserted into H the distribution of the children already visited after stopping, we would obtain 1.25. In that way we would count the stopping effect twice. The existing measurements are made with noisy readings. They stop at the first failure. Their distribution of visited children cannot directly replace the law of the required number.

The dependence on previous readings could also contribute to the difference. It is not, however, an isolated explanation of the entire deficit.

A population approximation was checked on the corrected version of the construction. It gives $S/N = 0.20719$, against 0.20700 in the recorded control. The final weight of the cell is 384, equal to the initial one. The means of spatial registers and memories are 0.94037 and 0.47909.

An earlier revision, 03, mistakenly described an older version instead of the corrected version. In that older version the weight was not conserved. This correction changes the reading of the result and is reported here. The approximation remains an aid, not an autonomous prediction. Close agreement in a single fraction does not prove general accuracy.

The proof above does not depend on this approximation. Nor do the completed-level tests of Appendix Q depend on it.

Appendix M. The history of the readings and where the memories accumulate

With the values A_r and B_r defined in Appendix I, we can calculate the probability of a new confirmation. The exact formula is:

$$\Pr(\text{confirm}) = B_r + (A_r - B_r)\Pr(x_i = x_j \mid \mathcal{F}).$$

Here $\mathcal{F}$ collects the history of the network that we can know. It also collects the information on how the link was chosen.

The formula uses the probability that the symbols at the two endpoints are equal, given that information. The probability can be calculated using the whole history of the network. The result can differ from the one obtained by looking at the confirmations and distinctions of a single link.

Consider an initial pair of independent symbols, with $r = 1/2$. The symbols are chosen among four possibilities. Before any reading, the confirmation probability is 0.25. One confirmation brings the probability of the next reading to 0.296875. In a triangle, instead, we look at the confirmations on the sides AB and BC. They bring the confirmation probability on side AC to 0.26171875. A direct count of all the cases proves both results. If $r = 0$, formula I1 always gives the exact probability $1/m$. This holds even after taking into account the whole previous history of the network.

In the recorded sample with a coherent region we recalculated the probability of 419 readings. We also used the common neighbours. The mean change of the prediction is only 0.00109. The mean absolute change is 0.00662 and the largest is 0.11412. Positive and negative changes cancel in the mean. So 0.00109 is not a bound on the error. The calculation does not include the longer paths in the network. Moreover, 45 of the 1,500 sampled readings were excluded because their neighbourhood was too large. It is a limited check, not a complete calculation of the probability based on the whole history.

To study where the unrealized distinctions accumulate, we use D_{stop}. It is the counter called U in the model's notes. It is not the physical energy U of Appendix D. For each distinction we measure the current distances in the network. They are the distances from the two endpoints to the initial region. We assign the distinction to the smaller distance. Then we divide the number of distinctions at each distance by the number of spatial registers at that distance. Let us take $N = 65536$ and the run with a coherent region. There the ratios at distance zero and one are 106.875 and 9.786 respectively. In the control, they are 4.25 and 0.245 [16].

Under this way of assigning the counts, the data show a strong accumulation near the initial region. They do not prove that the accumulation is confined exactly to two distance levels. Nor do they prove that the outer profiles are equal. At distance three, the ratios are 0.240 and 0.125. The same distance in network steps is not automatically the same physical radius. Moreover, a single pair of runs does not remove all the differences due to age, geometry or initial preparation.

These results are counts in an exploratory network. Further physics and further comparisons are needed to obtain a mass profile or a pressure field. They are also needed for a causal transport law or a gravitational lensing signal. From this accumulation we do not infer that it is dark matter. Nor do we infer an exclusive observational signal of MM.

Appendix N. How to assign energy without changing B17

We introduce K_E. It is an effective count to which energy is assigned in the exploratory model of space. It is not automatically the active count of the B17 model. We choose two positive constants, v and ε, and adopt:

$$V = vS, \qquad E_D = \varepsilon K_E, \qquad P_D = -\frac{dE_D}{dV}, \qquad w_D = -\frac{d\log K_E}{d\log S}. \qquad (N1)$$

In words, the volume V grows in proportion to the number S of spatial registers. The energy E_D grows in proportion to the energy count K_E. The pressure P_D depends on how the energy changes when the volume changes. The ratio w_D compares the growth rate of the energy with that of the spatial registers. It does so with the minus sign shown in the formula.

[IMPORTED EFFECTIVE POSTULATE] Formula N1 assigns physical volume and energy to counts of relations. It also assumes a uniform component, without other energy exchanges. The last relation follows from the work balance. It is obtained by dividing by the energy density $\rho_D = E_D/V$. To apply it to averaged counts, one must specify how the data are grouped and smoothed. It does not give an instantaneous pressure to each single random step.

The simplest choice is $K_E = S$. In this case the density stays constant and equals ε/v. The pressure-to-density ratio is $w_D = -1$. It is a possible microscopic interpretation of a constant-density component. It does not fix the observed value of that density. Nor does it account for the separate vacuum residue present in the inherited cosmology.

We can instead choose $K_E = S + \eta D_{\text{stop}}$. The number η is a constant, nonnegative relative weight. It applies to the energy assigned to the unrealized distinctions. In that case:

$$w_D = -1 - \frac{d\log(1 + \eta D_{\text{stop}}/S)}{d\log S}. \qquad (N2)$$

The formula says that the ratio depends on the relative amount of unrealized distinctions. It depends on how this amount changes as S grows. With $\eta = 1$ we calculate the logarithmic slopes between 32,768 and 131,072 total records. They are about -1.0242 in the control and -1.0844 in the coherent run. They are slopes over finite intervals for candidate energy counts. They are not measured cosmological equations of state. Moreover, a localized component cannot be described with the uniform formula N1 without further work.

Assigning energy to all the memory records M_{mem} is a different choice. D_{stop} is only a part of those records. If the memory exchanges energy with other matter, the balance also contains exchange terms. In that case the pressure cannot be derived directly from the balance without exchanges.

General relativity describes how the energy-momentum tensor acts as a source of geometry. It does not establish that a record has zero energy merely because it adds no spatial volume [15]. For MM, assigning energy only to S is a minimal effective choice. It is not a consequence of Einstein's equations. The assignment must remain explicit. The way in which energy couples to matter must also remain explicit.

This extension makes the question more precise. But it does not change the observational result already reported. The original B17 construction keeps the connection described in Appendices D-F. It is the connection between capacity and volume and between activity and energy. The parameters kept fixed in Appendices E and G remain unchanged. All the results of Table 1 also remain unchanged. The observational comparison does not validate the choices N1-N2. It does not decide which microscopic assignment is correct. The extension proposes a clearer mechanism and a conditional calculation of growth. It does not secretly modify the cosmology already compared with the data.

Appendix O. An explicit rule that depends on the state

We denote by η the complete state. It includes the ordered mesh and the symbol retained by each spatial register. It also includes the reading channels and the number of memories. In the internal calculation there is also an ordered list of the requests still to be checked. Each reading uses rule I1 and the actual symbols at the two endpoints. It need not reconstruct the probability of the symbols starting from a shortened record of past readings. If some symbols are not observed, averaging requires their complete conditional distribution.

[CANDIDATE MM RULE] In the uniform version, we choose a tetrahedron c of generation g with a weight proportional to 2^{-g}. A generation indicates the level of the subdivision. We resolve the oldest incompatible request first. If two requests have the same generation, we choose the one with the smaller identification number. We use the ordered subdivision into two parts described by the model's rule. Each successful interposition receives a new symbol chosen uniformly. Confirmations and failed distinctions produce the records defined in Appendix I. These choices specify the candidate process. The ontology does not impose them uniquely.

Consider a cell event that terminates. Consider all its possible histories of reading and of choosing the new symbols. We denote a history by h. Its probability is $P_\eta(h \mid c)$. It is the product of the probabilities of the single conditional readings and of the draws of new symbols. We call $R_S(h)$ and $R_M(h)$ the numbers of spatial registers and memories produced by that history. Then:

$$\Gamma_X(\eta) = \sum_c \omega_c(\eta) \sum_h P_\eta(h \mid c) R_X(h), \qquad X \in \{S, M\}. \qquad (O1)$$

In words, we want to find the mean growth of S or of M. We sum the results of every possible history. We multiply each result by its probability and by the weight of the chosen cell. The cell weights sum to one. Suppose that the event terminates with probability one. Suppose also that the number of records produced has a finite mean. In this case O1 gives the exact conditional mean of that number. In J2, the two growth coefficients are precisely these quantities. They are quantities that depend on the state. O1 defines an explicit mean. It is not a computational shortcut. It does not reduce the whole process to two variables.

We denote by $\mathcal{P}$ the complete rule. It includes the choice of the cell and the copying of a memory. The distribution of all the states evolves like this:

$$\mu_{k+1}(\eta') = \sum_{\eta} \mu_k(\eta)\mathcal{P}(\eta, \eta'). \qquad (O2)$$

The formula says how the probability of finding the next state η' is obtained. All the previous states are considered. The probability of each state is multiplied by the probability of passing from that state to η'. Then the results are summed. The elementary process of readings and requests is defined even before the results of an event are grouped. To use O1-O2 at the level of the grouped event, the termination conditions stated above are needed. The finite runs verify that every executed event terminates. They do not prove that every possible history of infinite subdivisions terminates.

Copying a memory leaves the mesh and the symbols unchanged. If we remove these copies, we obtain exactly the embedded chain tested in Appendix Q. This, however, does not preserve the growth exponents of the total number of records. Nor does it define physical time.

Appendix P. Why counts alone are not enough

Let us think of an exact rule that looks only at the number S of spatial registers and at the number of memories. We can show with a counterexample why the rule that depends on the state cannot be replaced with a rule of that kind.

We start from the same periodic mesh of $3 \times 3 \times 3$ cells. We use the same channels with $r = 1/2$ and the same counts. A periodic mesh connects opposite edges, as if the grid closed on itself. All the initial reference links are compatible. So their events require no additional checks.

In preparation A, all vertices have the same symbol. In preparation B, the symbol depends on the first index of the grid, taken modulo three. Each initial link joins opposite corners of the cube. In preparation B the symbols are different, also across the periodic edges. Both preparations are allowed among the four symbols of the model. For each preparation, the selected initial events have the same confirmation probability. Rule I1 gives:

$$\Gamma_S(\eta_A) = \tfrac{9}{16}, \qquad \Gamma_S(\eta_B) = \tfrac{13}{16}, \qquad (S, M)_A = (S, M)_B. \qquad (P1)$$

The two states have the same counts S and M. But they produce different mean rates of new spatial registers. So no exact rule that uses only those two counts can describe all the allowed states. It is a mathematical counterexample, not a small numerical imprecision. It does not exclude an approximate rule for a restricted group of states. Nor does it exclude adding other variables that summarize the relevant correlations. We do not claim that a universal reduced rule exists even for the set of random initial states considered.

We can also check the counting of events without a simulation. Take an initial request with two ordered checks, both at the ends. We denote by $\alpha_0, \alpha_1, \alpha_2$ the distinction probabilities of the three checks. There are four possible histories, because the process stops at the first confirmation. The first history is an immediate confirmation. The second is a confirmation at the first child. The third is a success of the first child followed by a confirmation of the second. The fourth is complete success. The pairs of outputs produced (spatial registers / memories) are, in order, $0/1$, $0/2$, $1/2$ and $3/0$.

We weight each result with the probability of its history. With these weights, the mean of spatial registers is $\alpha_0\alpha_1(1 + 2\alpha_2)$. The mean of the total number of readings is $1 + \alpha_0(1 + \alpha_1)$. The supplied check lists all four histories for 125 triples of probabilities. It also includes the triples with values zero and one. In the third history, the earlier successes remain recorded. Cancelling them all together at the end would be a different rule.

The result clarifies the structure of the model. The reading rule and the complete state determine the growth. The compressed descriptions have a precise task and an approximation error that can be measured. This is a complete mathematical formulation for the specified finite events. It does not derive the rule, the dimension, the energy unit or the cosmic clock.

Appendix Q. Network tests on completed levels

A second, compact version tests the uniform candidate process. It uses four symbols and channels with $r = 0$ or $r = 1/2$. For each channel four independent random seeds, 301-304, are used. A random seed fixes the initial sequence of random choices. Each run starts from a periodic mesh with 27 vertices and 162 tetrahedra.

The runs continue until the group of subdivisions denoted by $k = 2$ records all the required rings. They are 1,728 of type D, 5,184 of type F and 5,184 of type E. In all there are 12,096 distinct successful interpositions. All eight runs reach this result.

The means count the initial requests of the generation-2 group. They also count all the preliminary readings that those requests trigger. Those of older groups are included. The results are per cell event. They are not per spatial registration and not per interval of cosmic time. The four runs have the same weight in the mean. The brackets give the minimum and maximum value among the four seeds. They do not give a confidence interval.

Channel and measured quantity	Mean [range of the seeds]
Memoryless: spatial registers produced	0.92828 [0.92360, 0.93543]
Memoryless: memories produced	0.43216 [0.42798, 0.43514]
Memoryless: confirmation share	0.24947 [0.24848, 0.25163]
Halved signal: spatial registers produced	0.89297 [0.88915, 0.89569]
Halved signal: memories produced	0.45808 [0.44954, 0.46554]
Halved signal: confirmation share	0.26593 [0.26316, 0.26922]

Every cell event satisfies the record balance. The total weight assigned to the choice of cells stays unchanged. One check uses only geometry. It measures the preliminary requests and then restores the initial mesh and identifiers. It makes no random draws. The tests check the restoration and the averaging of the reading channel. They also check the completion of the counts and the fingerprints of the results. They do not prove that this version is equivalent to every earlier prototype. Here the rule for resolving ties is specified explicitly.

For each run we can use the formula of Appendix K. We insert into it the measured distribution of the preliminary requests and the observed confirmation share. Predicted means of spatial registers and memories are obtained. They are $0.92783/0.43078$ for the memoryless channel and $0.89387/0.45207$ for the halved signal. It is a check on the same run, not an autonomous prediction. In the case of the halved signal we can instead keep the old probability of one quarter. In that case the means become $0.92600/0.43025$. The old probability cannot be transferred unchanged.

Tests with larger initial meshes are also retained. Six of the eight runs stop at the declared cap of 80,000 spatial registers. They stop before completing the group $k = 2$. None is used as a completed group. Four seeds at a single size of the completed mesh do not constitute a test on infinite networks or on asymptotic behaviour. These runs do not include coherent regions or their formation. They do not include memory copies or energy assignments. They do not add comparisons with observational data.

Appendix R. Check of the pair approximation and present limits

The later approximation is based on pairs and clusters of links. It improves several geometric predictions [16]. For example, the reported fractions of preliminary requests for D, F and E are about $0.370/0.610/0.502$. The reference values obtained from geometry are $0.370/0.611/0.500$. It is useful evidence of the role of correlations. It does not validate the whole reading network of Appendix Q.

A further check verifies the four-entry tables used for each pair. Their probabilities for the single nodes must coincide with the marginal probabilities of the same nodes. These are calculated separately. A consistent family of probabilities for those variables must satisfy both conditions. The fact that each table sums to one is not enough.

The tables checked have nonnegative values. Their probabilities sum to one with a discrepancy of about one part in ten trillion. During active evolution, however, we look at the largest discrepancy between the pair probabilities and those of the single nodes. It is 0.010728 with 1,500 calculation steps and 0.010749 with 3,000 steps. The interval examined requires both single-node probabilities to stay between 0.05 and 0.95. The discrepancy remains even when the number of steps is increased. It is not due to rounding of the sum. Two resolutions, by themselves, do not prove a general convergence.

Consequently, these tables remain an approximation. They cannot be treated as a single exact model of joint probabilities that also has the reported marginal probabilities. To condition a reading on the known records of several neighbours, their joint probability is needed. Dividing separately by each marginal probability adds an independence assumption. It is not always equivalent to the exact calculation of the conditional probability. This limit does not invalidate the explicit rule of Appendix O. Nor does it invalidate the tests of Appendix Q.

The values of memory production based on pairs are 0.4531 and 0.4553. They differ by about 0.48 per cent. The difference is not smaller than 0.25 per cent. They are results of a comparison that transfers information along a chain. They are not a demonstrated measure of the accuracy with which memories are produced in a noisy network. To compare them, the groups of runs, the reading histories and the geometric averages must be matched.

The research materials include the English implementation of the state-dependent process. They include the eight complete run records and the larger tests stopped at the declared cap. They also include 39 reusable checks of algebra, counting and evidence. The 39 checks recheck the supplied results. A separate simulation regenerates the trajectories. The check of the pair approximation includes the fingerprints of the source code and a description of the method in English. The original Italian prototype is kept locally. But it is not included as a reusable English implementation.

Now the mechanism for finite events and its conditional mean are specified. The limited test on uniform networks is also specified. A reduced description that is always accurate and the physical calibration remain open. The formation of coherent matter and a unique derivation from MM's ontology also remain open. B17 and its observational comparison remain unchanged. This work is not exclusive observational evidence in favour of MM.

Appendix S. The model compared together with three kinds of observations

This appendix and the three that follow compare the relational construction together with three kinds of geometric measurements of cosmology. The parameters of standard cosmology can adjust to the data. The outcome is threefold.

The model obtains a good simultaneous fit. At the minimum of the profile likelihood, the data include 1,590 supernova light curves and thirteen measurements of baryon acoustic oscillations. They also include three summary numbers for the geometry of the cosmic microwave background. The total chi-square is 1420.05. The result is at the same level as that of the cosmological constant. No block of data is sacrificed to fit another. The sampled chain of Appendix T reaches 1420.02. The difference is numerical, not physical.

The values of the standard cosmological parameters come out of the comparison, instead of being imposed. The best fit of the profile gives $h = 0.678$, $\omega_b = 0.0226$ and $\Omega_m = 0.306$. Each is within about one standard deviation of the Planck 2018 values. The result holds within the adopted FLRW framework and description of recombination. The photon temperature, the neutrino content and the branching law are imposed. Beyond these, no other cosmological value is imposed.

The free amplitude of the relational component is also bounded. Until now it was a conventional choice. The parameter p measures how far the component departs from a cosmological constant. Now it stays below a few hundredths. This excludes the larger value kept fixed in the restricted comparison of Appendices G and H.

The construction also has a precise consequence. By law S1, the mean ratio $\bar{w}_D$ of the relational dark component always stays greater than or equal to -1. This holds whatever the constant part of the vacuum. It cannot enter the regime called *phantom*. In that regime the ratio falls below -1. The present data do not require such a crossing. The model fits well without it. A future release might establish a crossing with a decisive level of certainty. The level would be defined by the release itself. A release of that kind would exclude this form of the model. It is a consequence stated before knowing that result.

Appendices G and H compare a curve kept fixed with supernovae. The comparison is over a stated late interval. That comparison cannot reach the cosmic microwave background. The reason is structural, not practical. The solution starts

from the initial state $a_i = 1/2$. Continuing it towards the past, the growing component reaches its first zero around $z_* \simeq 1.66$. Beyond that point, the counting model admits no state. A construction that stops at a redshift of about one and a half does not reach the epoch of recombination.

Here we compare a different object. We use the same branching law. The law is switched on locally and at a late epoch by the gate function described in Appendix O. We follow the mean expansion from the seed of the network up to today. Before the switch-on, the dark component is inactive. The expansion is the standard one. The model thus has an ordinary early universe and an ordinary recombination. As a consequence it also has a characteristic distance of the sound waves and an acoustic scale. They are to be compared with the summary data. The gate function makes the comparison possible. It is not a device introduced to fit the data.

Three blocks of observations enter. We do not refit the single blocks and do not change their weight. The supernovae are those of Pantheon+ above redshift 0.01 [19]. They are used without the SH0ES calibration. They are 1,590 light curves of 1,473 distinct supernovae. The duplicate curves remain included. The published matrix of statistical and systematic uncertainties is used. A single common brightness shift is removed with an analytic calculation. It is not the DES sample of Appendix G. Its 1,448 objects have the same number as the Pantheon+ selection used in B18. That selection goes up to redshift one. But its objects are not the ones used here. The global comparison uses the whole redshift range of Pantheon+.

For the baryon acoustic oscillations we use the DESI DR2 summary measurements [9]. The cosmic microwave background enters only through the summary distance values of Planck 2018. They are the shift parameter R, the acoustic scale ℓ_A and the baryon density. They enter together with their correlations [17]. These three numbers summarize the geometry of the last scattering surface of light. They are not the probability calculated using the whole temperature and polarization spectrum. This distinction is particularly important for the present comparison. It will be taken up again in the limits.

The mean evolution is calculated in closed form. The count equation is linear and has a scalar coefficient. The solution is obtained by applying to the initial state the exponential of the integral of the gate function. From the standard densities one then derives, by integration, the dark fraction and the expansion rate. In the same way one derives the comoving distances and the characteristic distances of the sound waves. Recombination is recalculated with HyRec using the actual expansion history.

The approximate calculation concerns only the two redshifts of recombination and the blocks used for the certification. The declared errors enter the uncertainties, instead of being ignored. To check the approximate calculation, it is redone 128 times. Each time the point that it will then have to predict is removed. In 4,608 decisions no false acceptance occurs. Of these, 4,572 decisions fall within the certified interval. Every supernova row stays within one millionth of a standard deviation. Every row of the baryon oscillations stays within 0.24 per cent of a standard deviation. Every row of the cosmic background stays within 0.005 per cent. In the certified interval 4.1 per cent of the cases that should have been accepted are rejected. It is a cautious error, because it rejects cases instead of accepting invalid ones. The certification holds only in the declared interval.

$$\bar{w}_D = -1 + s(u)\, p\, f(u), \qquad f = \frac{m_B}{m_A + m_B}. \qquad (S1)$$

Here s is the gate function. It is zero before the switch-on and one after. The symbol f is the share of the active registers that are of type B. The symbol p is the branching probability. The formula shows that the data do not bound only p. They bound the amplitude of the whole product. A small value of p reduces this same quantity. A late switch-on also reduces it. A branching component that makes up only a part of dark energy also reduces it.

Appendix T. What the data bound

In brief	Value
Upper bound on p, one-sided at 95 per cent	0.0724, from the converged sampled chain
Lowest total chi-square of the sampled chain	1420.02: supernovae 1403.87; baryon oscillations 10.98; cosmic background 5.17
Minimum of the profile likelihood	1420.05: supernovae 1403.4; baryon oscillations 11.6; cosmic background 5.1

Best fit of the profile	$h = 0.678$, $\Omega_m = 0.306$
Medians of the chain	$h = 0.6764$, $\Omega_m = 0.3066$
Cosmological constant, same calculations and same data	1421.19
Main methodological limit	the cosmic background enters with three summary distance values, not with the whole temperature and polarization spectrum

The three transition parameters $(p, u_c, \Delta u)$ and the three parameters of standard cosmology (h, ω_b, Ω_m) are sampled. The two spatial parameters of Appendix U are also sampled. The first three describe the dark part that is free to vary. They are the branching probability, the epoch at which the gate function opens and its width.

The model does not leave a function of redshift free. It does not build $w(z)$ by adding powers or by using a series of nodes. It does not reconstruct the history separately. Once the three parameters are fixed, S1 and the count equation determine the whole history, including recombination. A comparison of this kind can fail in a way in which a flexible parameterization cannot fail.

To show the result we first use a *profile*. We fix a value of p. Then we look for the minimum chi-square by varying the standard cosmology and the gate function. It is a direct way to check what happens at each value of p. It is done before summarizing a probability distribution.

p	Best total chi-square	Difference from the minimum
0.25	1468.49	+48.44
0.20	1451.94	+31.88
0.15	1437.64	+17.59
0.10	1426.77	+6.72
0.05	1420.75	+0.70
0.03	1420.05	0
0.01	1420.51	+0.46

Each row was also optimized with respect to the gate function. The late switch-on is mildly preferred. With $p = 1/4$ it is worth about six units of chi-square against the two earlier switch-on epochs tested. The width of the function is worth about one unit. The order of the values of p is the same at every epoch, because the observations of the expansion respond to the product in S1. They do not respond only to the moment at which the function switches on.

The minimum gives an acceptable fit. The contributions to the chi-square are 1403.4 for the supernovae, 11.6 for the baryon oscillations and 5.1 for the cosmic background. It is obtained with $h = 0.678$, $\omega_b = 0.0226$ and $\Omega_m = 0.306$. The one-parameter interval corresponding to $\Delta\chi^2 = 3.84$ closes near $p \simeq 0.08$.

The reference comparison uses the same calculations. We set the vacuum fraction of equation (8) equal to one. The branching component disappears and a cosmological constant remains. We use the same closed solution for the expansion and the same calculation of recombination. We use the same summary distance values, in the same convention. We also use the same three data blocks. The cosmological parameters, too, can adjust freely. The result is 1421.19, with $h = 0.6837$, $\omega_b = 0.02253$ and $\Omega_m = 0.3014$.

The cosmological constant has three free parameters. The relational model has six. The minimum of the relational profile

is 1420.05. It is about one unit of chi-square below the reference, on 1,606 measurements. But the dark sector uses three more free numbers than the cosmological constant. The cosmological constant has none. An improvement of one unit is not a gain per parameter. For this reason the criteria that reward simpler models favour the cosmological constant. The comparison provides no evidence in favour of the relational construction over a constant.

The comparison, however, also shows a distinct result. A dark component derived from a counting law meets the reference, using the same data and the same conditions. This component has no freedom to choose the shape of its history. It meets the reference at the same time for supernovae, baryon acoustic oscillations and the geometry of the last scattering of light.

Let us take the value $p = 1/4$ of the curve kept fixed in Appendices G and H. When the three data blocks are combined, it is disfavoured by about 48 units of chi-square. In the first table, the exact difference between the minimum and the row $p = 1/4$ is 48.44. The worsening is shared among the blocks. It does not come only from the last one added. From the minimum to $p = 1/4$, the chi-square of the supernovae goes from 1403.4 to 1418.8. That of the baryon oscillations goes from 11.6 to 25.1. That of the summarized cosmic background goes from 5.1 to 24.6. The preferred expansion rate falls from $h = 0.678$ to 0.643. This result concerns that configuration kept fixed. It does not refute the branching law. The model can reduce the amplitude of S1 in another way too.

The other way is the vacuum fraction ν_v of equation (8). It indicates which part of dark energy is a genuine constant, instead of the branching component. For the rest of the work it is fixed at zero. This is the strong reading, according to which all of dark energy is relational. We can let it vary, keeping the gate function at the middle epoch of the previous table. In that case we obtain:

ν_v	$\Delta\chi^2$ **with** $p = 1/4$	$\Delta\chi^2$ **with** $p = 0.03$
0	+54.04	0
0.50	+13.52	+0.07
0.80	+1.75	+0.44
0.90	+0.52	+0.62
0.95	+0.47	+0.72

The differences in the second table are calculated with respect to its own minimum. The minimum is reached with $\nu_v = 0$ and $p = 0.03$, keeping that gate function fixed. These differences are not directly comparable with the first table. The rows of the first table were also optimized with respect to the gate function.

With $\nu_v = 0.90$, the product $(1 - \nu_v)p = 0.025$ gives the same chi-square obtained with $p = 0.03$ and $\nu_v = 0$. Beyond that fraction, the two columns coincide. The model has become a cosmological constant and the data no longer bound p. The result must therefore be expressed like this:

$$(1 - \nu_v)\, p \lesssim 0.03 \text{ to } 0.07. \qquad (T1)$$

It is a bound on how far the relational component can depart from a cosmological constant. It takes into account its share of dark energy. It is not a bound on the share itself. In the strong reading, $\nu_v = 0$: all of dark energy is relational and T1 bounds p. In the weak reading, the model can have $p = 1/4$. But it must attribute nine tenths of dark energy to an ordinary constant. That is a theory different from the one proposed in this article.

The sampled chain answers the question about the distribution of p. It reached the declared criterion for considering it stable. It used 32 moving sampling points. It also used a rule that explores different combinations of the parameters together. It reached 8,000 steps. The largest autocorrelation time is 103 steps. In other words, the sequence retains part of its own memory for about 103 steps. The chain is 77 times longer than this time. That is much more than the 50 times required. Moreover, between checks, every autocorrelation time changed by less than 2 per cent.

We discard the first five autocorrelation times and thin the results. 8,896 samples remain. About 2,300 of these are

independent. In the strong reading, the one-sided upper bound for p is 0.043 at 68 per cent. It is 0.072 at 95 per cent and 0.097 at 99 per cent. The median is 0.033. The lower edge chosen in advance for p is instead 0.01.

The values of standard cosmology come out as $h = 0.6764$, $\omega_b = 0.02258$ and $\Omega_m = 0.3066$. The respective 68 per cent intervals are $[0.6714, 0.6806]$, $[0.02246, 0.02271]$ and $[0.3022, 0.3114]$. The switch-on epoch, its width and the two spatial parameters cover all the intervals allowed at the start. The comparison does not determine when the transition takes place. It only bounds how large its amplitude can be.

The lowest total chi-square reached by the chain is 1420.017. The contributions are 1403.87 from the supernovae, 10.98 from the baryon oscillations and 5.17 from the cosmic background. It is close to the minimum of the profile in the first table, 1420.05. The two values, however, come from different calculations. The profile uses the closed solution directly. The chain uses the certified approximate calculation.

The best point of the chain falls between the points tested in the profile grid. Its gate function starts a little later and is a little narrower. This accounts for the difference of 0.03 in the chi-square. The difference is numerical, not physical. The bound on the marginal distribution of p and the interval of the profile answer different questions. So they are reported separately.

Two other chains give compatible bounds. They are not, however, added to the main chain to give more weight to the result. The first uses the same certified approximate calculation and a standard sampling rule. It stops at 11,500 steps. It reaches 42.7 autocorrelation times, fewer than the 50 required. Its bound at 95 per cent varies from 0.073 to 0.076. It depends on the initial stretch discarded and on the thinning. The other is a preliminary test of the method. It uses twelve of the 128 certification points and 3,000 steps. It gives a bound of 0.072. Both are documented in the research materials.

Why the bound is physically informative

A bound indicates where the model can lie. The agreements among the data are what keep it in that interval. They have the same weight in the result.

First, at the minimum the three data blocks agree with one another. The supernovae give 1403.4 for 1,589 degrees of freedom. The baryon oscillations give 11.6 for 13. The summarized cosmic background gives 5.1 for 3. With the standard parameters free, no block pulls in the opposite direction to the others.

Second, at the best fit of the profile, $h = 0.678$, $\omega_b = 0.0226$ and $\Omega_m = 0.306$ are within about one standard deviation of the Planck 2018 values. The three data blocks share no systematic errors. The result is obtained with the transition present.

Third, the data show exactly the degeneracy predicted by formula S1. The vacuum fraction and p trade against each other along the product $(1 - v_v)p$. The agreement reaches the second decimal place. It is a prediction of the model's structure that the data confirm. It is not a feature added to fit them.

Fourth, the branching law forbids crossing the limit $w = -1$. The best fit does not require that crossing.

Fifth, when the data indicate a preference for the epoch of the transition, they go in the expected direction. It is the direction expected for the picture of the local transition. The data favour a late switch-on. The indication is worth a few units of chi-square for large values of p. At the minimum it is worth less than one unit. We record it, without building further conclusions on it.

None of these results is a detection. Taken alone, each is modest. Together, they show a concordance among the observations. It is information that the bound alone would not give. We report it with the same prominence as the bound.

What the comparison covers

The cosmic background enters as three summary numbers. It enters in the convention used when those numbers were derived. This convention reproduces the published shift parameter with a discrepancy of 0.03. It reproduces the acoustic scale with a discrepancy of 0.6 standard deviations. It differs instead by about three standard deviations from

an angular scale. That scale is calculated directly with a Boltzmann physics code. We use the published convention, without mixing it with another.

The geometric compression does not cover the late integrated Sachs-Wolfe effect. This effect can be tied to a transition that begins at low redshift. We do not calculate the temperature or polarization spectrum. The comparison is restricted to the interval covered by the certificate of the approximate calculation: $p \le 0.3$. We claim no results outside that interval.

Appendix U. When the noise is negligible and where the threshold lies

The expansion history describes a mean value. The gate function acts locally. So different regions switch on at different moments. Moreover, every coherence cell contains a finite number of records. So the mean value can also fluctuate. A coherence cell is a region treated as a group. To picture it, let us think of the seats of a classroom in which the pupils answer together.

The analysis does not simulate these fluctuations. For each sample it calculates an upper bound on the chi-square that could be due to the count noise. The count noise is the random variation produced by the finite number of records. The sample is used only if the bound stays below a fixed threshold.

The variance at the moment of switch-on is calculated exactly:

$$\mathrm{Var}(\delta \ln N) = \frac{2 - (a_{\mathrm{seed}}/a_{\mathrm{on}})^3}{N_{\mathrm{seed}}}. \qquad (U1)$$

The formula compares the fluctuation of the logarithm of N with the number of records present in the seed. The symbols a_{seed} and a_{on} denote the scale factor at the seed and at switch-on. With the other values unchanged, the variance decreases as one divided by N_{seed}. The more records there are in each cell, the smaller the fluctuation.

At a reference point, $N_{\mathrm{seed}} = 10^{20}$ and the coherence length is 100 Mpc. An Mpc, or megaparsec, is a large unit of astronomical distance. At that point, the bound on the chi-square is 4×10^{-11}. It is eleven orders of magnitude below what the three data blocks can resolve.

That point is a corner of the initial region explored by the chain. It is not an edge of the allowed values. The initial interval extends to $N_{\mathrm{seed}} = 10^4$. 53 per cent of the retained samples have a value below 10^{20}. The smallest value is about 1.7×10^9. The certificate keeps all these samples in the negligible-noise branch. It is not the initial choice of the interval that keeps them. The two spatial parameters are not bounded by the comparison. The data, then, are not preferring a finer network.

Let us read formula U1 the other way round. We obtain the only quantitative target offered by this work for the non-uniform sector. The omitted contribution reaches a hundredth of chi-square with about 4×10^{11} records per cell. It reaches a tenth with about 4×10^{10} and one unit with about 4×10^9. The declared threshold is crossed near 10^{11}.

These values hold for the reference coherence length, 100 Mpc. They change as the other parameters vary. For this reason the certificate checks each sample separately. The interesting interval is wide, down to 10^4 records. It lies mostly below the branch used here.

The analysis does not cover that noisier branch. The program rejects those samples instead of extending the results beyond the limit. If the noise is not negligible, the full covariance matrix across the data blocks is needed. Its determinant is also needed. A simple bound on the omitted contribution is not enough.

The program applies the same rejection when the count noise could leave traces beyond the declared limits. These are traces in the temperature of the cosmic microwave background or in its gravitational lensing. Three summary distances alone cannot absorb those traces.

In the 256,064 evaluations of the main chain, 12,433 samples were rejected for this reason. Another 791 were rejected because they exceeded the threshold of the omitted contribution. Another 168,273 were outside the initial interval of the parameters. This happens often with the sampling method used. In 333 cases the calculation was outside the interval foreseen by the emulator. Four cases failed the final check on the closure of the gate function. The chain summary keeps these counts.

So a lower bound shown by the chain for the number of records per cell marks the declared boundary of validity. It is not a measurement, whatever its value. The next step is to build that second branch. Then one must ask whether a network coarse enough to produce observable fluctuations is compatible with the same three data blocks. It is a defined task, not a forecast of what we will find.

Appendix V. A test fixed before the calculation for the non-uniform sector

This appendix sets out in advance what will be calculated and how the outcome will be read. It also sets out what will not be changed afterwards. The protocol is dated and is archived together with the other materials. The record precedes the result.

What is put to the test

The local transition produces noise in the counts. In Appendix U, that noise is negligible above about 10^{11} records per coherence cell. This holds at the reference coherence length. The test concerns a network coarse enough to make that noise important. It asks whether the three data blocks of Appendix S are compatible with a network of that kind.

The test considers the whole interval allowed by the model: from 10^{4} to 10^{25} initial records per cell. The number of initial records is denoted by N_{seed}.

How it is calculated

For every design point, the certificate stores a linear response. It is the response of every data row to each unit column of the count noise. A linear response indicates how much a row changes when an input changes. It is like noting how much each column of a sheet moves if one piece of data is modified.

The responses are stored for three variances: 10^{-4}, 2×10^{-4} and 10^{-3}. The closure coefficients are also archived. These coefficients say how much each row shifts on average per unit of variance. The second branch uses these values in every sample. With them it builds the mean shift $\mu(\theta)$ and the noise covariance.

In this calculation, the bound on the omitted contribution of Appendix U is replaced by the full form:

$$-2\ln\mathcal{L} = (r-\mu)^{T}(C + A\Sigma A^{T})^{-1}(r-\mu) + \ln\det(C + A\Sigma A^{T}). \qquad (V1)$$

The formula compares the residuals corrected for the mean shift with the overall uncertainties. Here C is the published covariance of the three data blocks. The symbol A is the stored linear response. The symbol Σ is the covariance of the count noise. Its scale follows U1.

The matrix inside the brackets includes the uncertainties of the data. It also includes those due to the count noise. The term with the determinant also takes into account how large this overall uncertainty is. The rank of the contribution $A\Sigma A^{T}$ does not exceed the number of noise columns. To calculate the inverse and the determinant, the Woodbury identity is therefore used. It is an algebraic method that simplifies the calculations when the added contribution has few independent directions.

The emulator supplies A and μ in the same cases in which it now supplies the rows of the certificate. The declared error of the emulator enters V1, as it already enters the present likelihood.

The rule for deciding

The chain uses the same initial interval as the present test and does not change it. The strong reading is used, $\nu_{v}=0$: all of dark energy is attributed to the branching component.

The marginal posterior distribution of $\log N_{\mathrm{seed}}$ is compared with a reference. A posterior distribution summarizes which values remain plausible after the data have been considered. The comparison is necessary because a bound set at the start can exclude the lowest values even without information from the data.

The reference is a second chain with the same sampling method, the same interval and the same rejections. But it has a

constant likelihood. In this way, a cut or a rejection near the lower edge removes samples from both chains. It is not mistaken for a measurement.

F is the name given to the lowest tenth of the allowed interval: $4 \le \log_{10} N_{\mathrm{seed}} \le 6.1$. For each chain, the fraction of samples that falls in F is turned into an effective count. To do so, the larger of two autocorrelation times is used. One is that of $\log N_{\mathrm{seed}}$. The other is that of the indicator that equals one inside F and zero outside.

The autocorrelation time measures how long the samples stay tied to the previous ones. The effective counts take this tie into account. For the two fractions, one-sided Clopper-Pearson bounds at 97.5 per cent are calculated. The bound U is the upper bound of the fraction in the data chain. The bound L is the lower bound of the fraction in the reference chain. They are statistical bounds used to compare proportions. If the data chain does not visit F, the value of U still stays above zero.

When effective counts are used, these bounds are approximate. They do not guarantee exact coverage for correlated samples. The rule is archived as a program (script). The program will be applied, not this paragraph.

Outcome not assessable

The rule cannot be applied if either of the two chains has fewer than 500 effective samples. Nor can it be applied if the reference chain has fewer than 50 effective visits to F. The chains can be extended once, up to four times the initial length. If even then the amount of data remains insufficient, the check is inconclusive on that side. It is not an outcome in either direction.

First outcome

If $U/L < 0.1$, the result gives a ratio greater than ten to one against a coarse network. It is conventionally strong evidence. In this case the one-sided 95 per cent lower bound of N_{seed} is reported as the result.

Second outcome

If the threshold is not reached, one does not conclude that the data say nothing about the lower part of the interval. The one-sided 95 per cent lower bounds are reported and compared anyway. They are those of the data chain and of the reference chain. But the test does not present a lower bound as its own result.

The upper side is also read separately, using the highest tenth. If the data indicate a coarse network, the next step would be to look for it in the matter power spectrum. The power spectrum is the way in which density variations change with size. To calculate it, coupled Boltzmann simulations would be needed. These simulations follow the cosmological components together.

Checks of the rule

Before archiving and again before the test, six synthetic cases with correlated chains are checked. The cases and the expected results are fixed and archived.

A constant likelihood produces the second outcome. A likelihood that vanishes below $\log_{10} N_{\mathrm{seed}} = 8$ produces the first. A chain with no samples in F and a chain that is too short both turn out not assessable. A posterior probability of 0.001 in F produces the first outcome. A probability of 0.02 moves the lower bound from 5.1 to 6.7. But it does not reach the threshold and produces the second outcome. All six cases pass the checks. If a repetition before the test did not pass them, the rule would not be applied.

Third outcome

The linearity test of the closure may exceed the 5 per cent tolerance even in a single row examined. In that case the branch cannot be certified with this method. The same holds if the emulator error on A exceeds one tenth of the standard deviation of the corresponding row.

In either of these cases, it is reported that the branch cannot be certified with this method. The two tolerances are not

loosened to obtain an outcome.

What stays fixed

After the result has been seen, the initial interval is not changed. No data blocks are added or removed. The gate function, the vacuum fraction and the thresholds stay as declared here.

The weak reading, with $\nu_v > 0$, is a secondary analysis. It will be labelled as such. It will be added to the materials in a dated revision, whatever its outcome.

Two consequences already stated by the model

Independently of this test, the branching law imposes $\bar{w}_D \geq -1$ at every epoch. This is established by formula S1 and by the averaging with any constant fraction of the vacuum. The model cannot enter the phantom regime. In that regime the ratio w falls below -1.

A future release on baryon oscillations or on supernovae might establish a crossing of that threshold. It might do so with decisive certainty, according to the criteria declared by the release itself. A result of that kind would exclude this form of the model.

Moreover, in the strong reading Appendix T bounds p, that is, the departure of the relational component from a cosmological constant, to a few hundredths. A larger measured departure would also exclude it, if it kept the pressure-to-density ratio above -1, as S1 requires.

The method

The author prepared the protocol, the programs and this text with the help of an AI assistant. The assistant wrote and ran the programs under the author's direction. This information serves to make the assistance part of the method, subject to the same reproducibility checks as the other steps. It is not a source of authority in one direction or the other.

Every number in this package was produced by a program. The test will be assessed by the programs, not by the author or by the assistant.

Glossary

Acceleration of the expansion. Increase over time of the speed at which cosmic distances grow. It is not a push of wind dragging the galaxies.

Acoustic scale. Characteristic size left by sound waves in the young universe. The summary data use it to study cosmic geometry.

Autocorrelation. Connection between a sample and the nearby ones in the same chain. If the chain has memory for many steps, nearby samples give less independent information.

BAO, or baryon acoustic oscillations. Trace left by ancient sound waves in the distribution of matter. Its size helps to measure cosmic distances.

Capacity. Total number of slots, including those inactive for the count being tracked. The connection between capacity and volume is an additional physical assumption.

Chi-square. Number that summarizes the differences between calculations and data, taking into account the uncertainties and their correlations. With the same procedure, a lower value indicates smaller overall differences.

Coherence cell. Region treated as a group when the fluctuations of the counts are calculated. The number of records in each cell helps to establish how much the noise matters.

Comoving. Following the mean expansion of the universe. A comoving region keeps its identity while its physical volume changes.

Count noise. Random fluctuation due to the finite number of records. In general, more records per cell mean smaller relative fluctuations.

Covariance. Table that summarizes the uncertainties and how much the errors of different measurements are related to one another. It is like a chart showing which marks tend to go up or down together.

Credibility at 95 per cent. Degree of compatibility assigned to the values after applying the model and the analysis. A 95 per cent bound is a boundary, not an exact measurement of the parameter.

Dark energy. Name given to the component used to describe the acceleration of cosmic expansion. It is not observed directly: its effects are studied.

Dark matter. Name given to the component used to describe some gravitational effects, for example in galaxies. It is a problem different from dark energy.

Dark sector. Part of the model that includes dark energy and any associated parameters.

Distal component. Component of the model tied to the links accumulated between epochs being compared. It does not indicate a greater energy due to the distance from the observer.

Dominant regime. Part of the evolution in which the most persistent growth mode prevails. It need not coincide with the entire observed interval.

Effective bridge. Set of assumptions and physical laws used to turn the counts of the model into observable quantities. It is not a fundamental derivation of all those laws.

Emulator. Approximate calculator that imitates a longer calculation after being checked. Here its uncertainties are included in the analysis.

Expansion rate. Measure of how quickly cosmic distances grow at a given moment. Having a large final growth does not say, by itself, how much time it took.

Fingerprint. Code calculated from a file to check whether it has been modified. It works like a short fingerprint of the content.

Fractionation. Attenuation of the comparison transmitted by a link. q measures the retained correlation; it is not the probability that a photon survives.

Gate function. Rule that describes when a component begins to contribute. In this test it can open at a different epoch and with a different speed.

Interposition. Insertion of a register between two others, so that the comparison passes through a new step.

Likelihood. It measures how well a model describes the data under a statistical procedure. The chain uses it to compare different choices of the parameters.

Matter power spectrum. Description of how large the variations in the density of matter are at different scales. In the future it could help in looking for the effects of a coarser network.

Memory-space extension. The separate candidate model of Appendices I-R: spatial registers and memories have distinct roles. Its reading channel r is not the original epoch channel q.

Metric. Rule that makes it possible to measure distances and times. A length defined by the model of links is not automatically a relativistic metric.

Mpc, or megaparsec. Very large unit of astronomical distance. It serves to describe separations between cosmic regions, not everyday distances.

Negative pressure. Pressure that, in the adopted cosmological equations, can contribute to the acceleration of the expansion. It is not a wind that pushes the galaxies.

Parameter. Number that can be varied in a model. Changing a parameter can change the predictions.

Parameter kept fixed. Value retained from the previous analysis. It does not mean that it was derived from the theory or that the new test was blinded.

Posterior values. Values of the parameters regarded as compatible after considering the data and the initial assumptions. They are summarized, for example, by the median and by the bounds of a chain.

Preregistration. Description, archived before the calculation, of what will be done and of how the outcomes will be read. It helps to make visible the choices made in advance.

Pressure-to-density ratio. The ratio $w = P/\rho$, where ρ denotes energy density. It expresses the contribution of pressure relative to density in the cosmological bridge.

Profile likelihood. Method that fixes a parameter and looks for the best possible chi-square by varying the others. It shows how the agreement changes for each fixed value.

Redshift. Relative increase of the wavelength of light during its journey. If the wavelength doubles, $z = 1$ and the stretching factor is $1 + z$.

Redundancy. Outcome that, in channel B, does not increase the active total. Its probability p is different from the share of correlation q.

Register. Element capable of carrying a distinguishable state, like one of the two signs on a square. It requires neither a conscious memory nor a human observer.

Relational link. Connection that relates distinguishable states, like two different signs on the blackboard. It is the starting element proposed by MM.

Sampled chain. Sequence of parameter values used to explore which choices agree with the data. It is like checking many possible combinations, instead of trying only one.

Scale factor. Number a that describes how distances change between references that follow the expansion. Their distances in comoving coordinates remain constant. In three dimensions, volume grows as the cube of a.

Slot. Place in which the model can record a state. It can be active or inactive for the count considered.

Standard deviation. It measures how much the values are spread around a mean or an estimate. The more standard deviations separate two values, the greater their distance compared with that spread.

Type Ia supernova. Stellar explosion whose light, after corrections and standardization, helps to estimate distances. Supernovae are not all identical light bulbs.

Vacuum residue. Constant part of the energy density kept distinct from the relational component. In this work its value remains unexplained.

How the work was done

The record MM 2.32 is a separate and restricted contribution with respect to the MM 2.30 unified volume. That volume is English edition 0.19, DOI 10.5281/zenodo.22730712. The record MM 2.30 has not been modified.

The bridge between capacity and volume comes from the working notes R30 and B14 of September 2026. The probabilistic realization of the link also comes from there. The rule with counts $2n + 1$ and one was developed in the first edition of the article. The same holds for the joint process of Appendix B. The branching family and the robustness region come from the research manuscript B17-R1 of 13 September 2026. The structural checks also come from there. That manuscript accompanies the record. It is not an independent observational confirmation.

Table 1 and the checks of Appendices G-H take up a report on the DES comparison dated 13 September 2026. They

have not been recalculated for this edition. The local procedure was fixed on 13 September at 04:56:05 UTC. It was fixed before acquiring the individual data and calculating their new likelihood. The published DES results as a whole were already known. The report also retains the DES selection with survey code 10. It also retains the comparison on the Pantheon+ sample and the related checks. The wider Pantheon+ sample of Appendices S-U is a distinct analysis. It has its own calculation and two further data blocks.

The sources of the DES-SN5YR and Pantheon+ data can be identified through the public commits c9a4fcafc4cbd19bd750dee47fc76194a45c181f and c447f0f. The Planck 2018 summary data follow the convention of the work that published them. The tests for the memory and space network derive from internal MM notes. The notes concern the distinction rule, the growth coefficients and the later approximations. The original notes and prototypes are credited to Claude. The English synthesis was developed with AI assistance under the author's direction. The same holds for a new implementation of the reading process and for further checks.

The separate research materials include editable texts, figures and internal mathematical checks. They also include results and the files of the cited observational data. They further contain file fingerprints, complete run records and reusable checks. For the process of Appendices O-Q, an English implementation, the results and the instructions to regenerate them are available. The check of the pair approximation includes a description of the method. The original Italian prototypes are not included as portable English programs.

The statistical check of Appendix S repeats the verification of the emulator 128 times. Each time it removes the point it has to predict. Out of 4,608 decisions no false acceptances are observed. Of these, 4,572 fall within the certified interval. The package also contains 39 checks of algebra, counting and evidence. These checks recheck the supplied results. A separate simulation instead regenerates the trajectories. The fingerprints of the materials help to verify that the files checked are the archived ones.

The complete comparison of Appendices S-U uses the background solution and the calculation of recombination with HyRec. It also uses an emulator based on Gaussian processes, its certification and the sampled chains. The files of the data tables and the original training materials are not included. So the archived package is not presented as a complete and standalone observational reproduction. A separate kit was prepared and tested with this edition. It collects the code needed for the profiles of Appendix T. It also collects the programs that download the three data blocks and HyRec-2 from the original sources. The kit is not part of this deposit. A separate record that will cite it is planned.

The documented commands repeat the mathematical checks of the formulated models. They do not prove the ontological origin of the rules. Mathematical proofs and numerical checks have different functions. The latter do not replace the former. Nor are they equivalent to an independent scientific review. The dated research documents keep their original scope. The preparation of this edition does not add a new fit to those documents. The S-U comparison is presented separately, with its own calculations and limits.

The original English article, of 29 pages, translated a refined Italian version. The English revision preserves equations, cosmological parameters, samples and observational results. It adds the new narrative and Appendices I-R. The local Italian edition does not yet contain those additions. For the package, the English text is the reference edition. The parity checks between translations concern only the earlier common content.

The observational comparison of Appendices S-U was carried out by the author with AI assistance. It is not presented as an independent scientific validation. The ontological architecture is proposed by the author. Formalization and writing were developed with AI assistance under his direction. Translation and editorial preparation also used an AI assistant. It is not claimed that a second research group took part. Nor is it claimed that there was an independent scientific review. Author identifier: ORCID 0009-0008-7618-9635.

This English text is a translation of the Italian student version, which the author read and approved.

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Cosmology, data and neighbouring programmes

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Model provenance and supporting materials

[11] Ugo Lanciano, MM 2.30 unified volume, English edition 0.19, source record identified by the author: DOI 10.5281/zenodo.22730712. This article does not modify that record. The 2.32 package is a separate, restricted contribution, not a replacement for the unified volume.

[12] Working notes R30, *Candidate bridge between MM and the distal hypothesis*, and B14, *Links, counting and regularity* (September 2026; titles translated from Italian). They supply the capacity-volume bridge and the probabilistic realization of a link, respectively. The channel with counts $2n + 1$ and one, and the joint process in Appendix B, were developed in the first edition of this article.

[13] B17-R1 research manuscript, *Robust distal growth from a relational branching model* (13 September 2026). Source of the branching family, robustness region and structural checks used in the appendices. It is not independent observational confirmation. The 2.32 package retains it as a companion manuscript.

[14] Research report, *A restricted observational transfer test of the MM distal background* (13 September 2026). Source of the frozen parameters, DES selection, Table 1 and diagnostics in G-H. Its results are inherited here, not recalculated: the independent comparison of Appendices S-U uses a different supernova sample, Pantheon+ above redshift 0.01 [19], with its own background, its own likelihood and two further data blocks. The companion PDF reproduces the dated report with a packaging note.

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